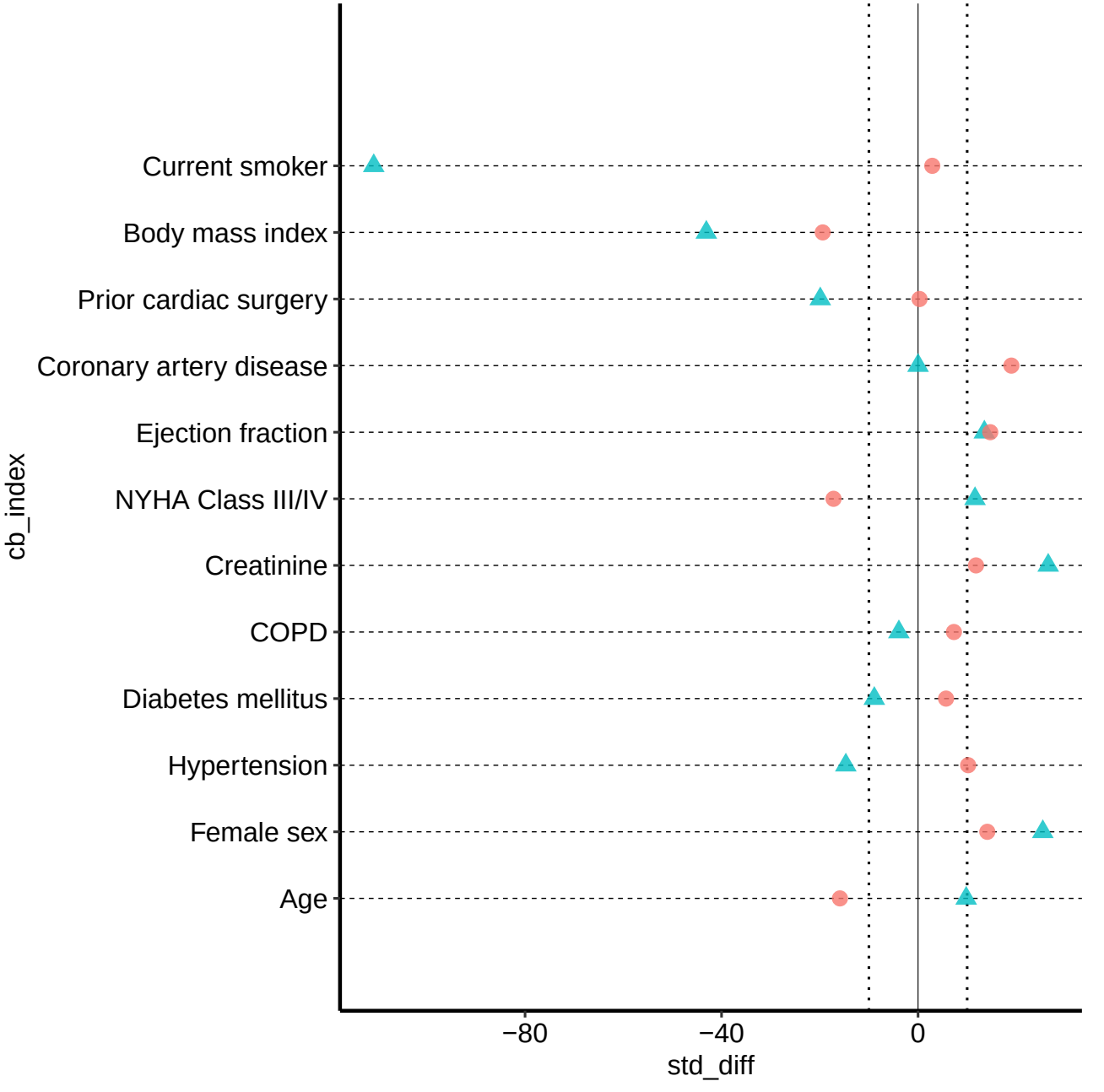
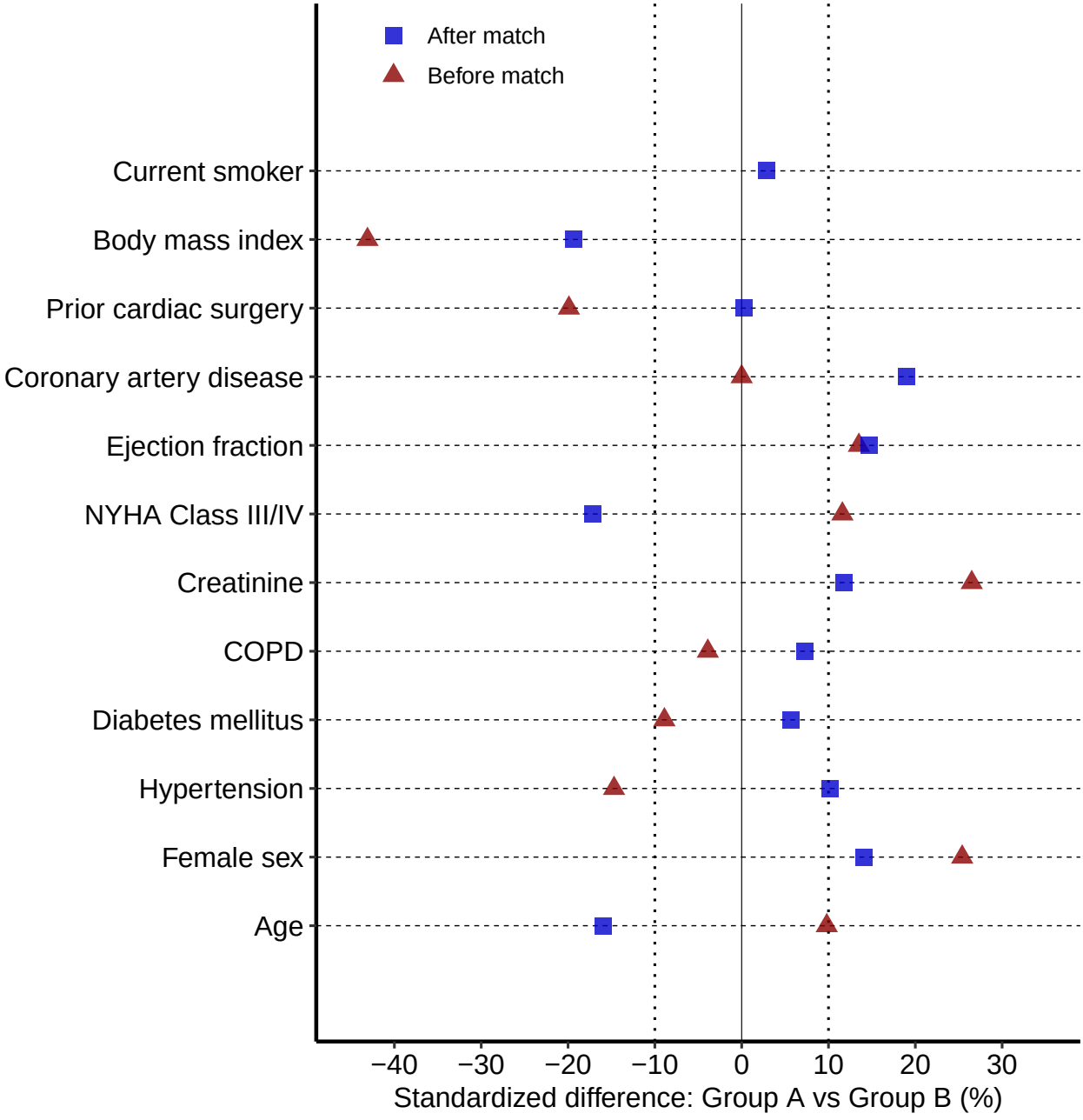
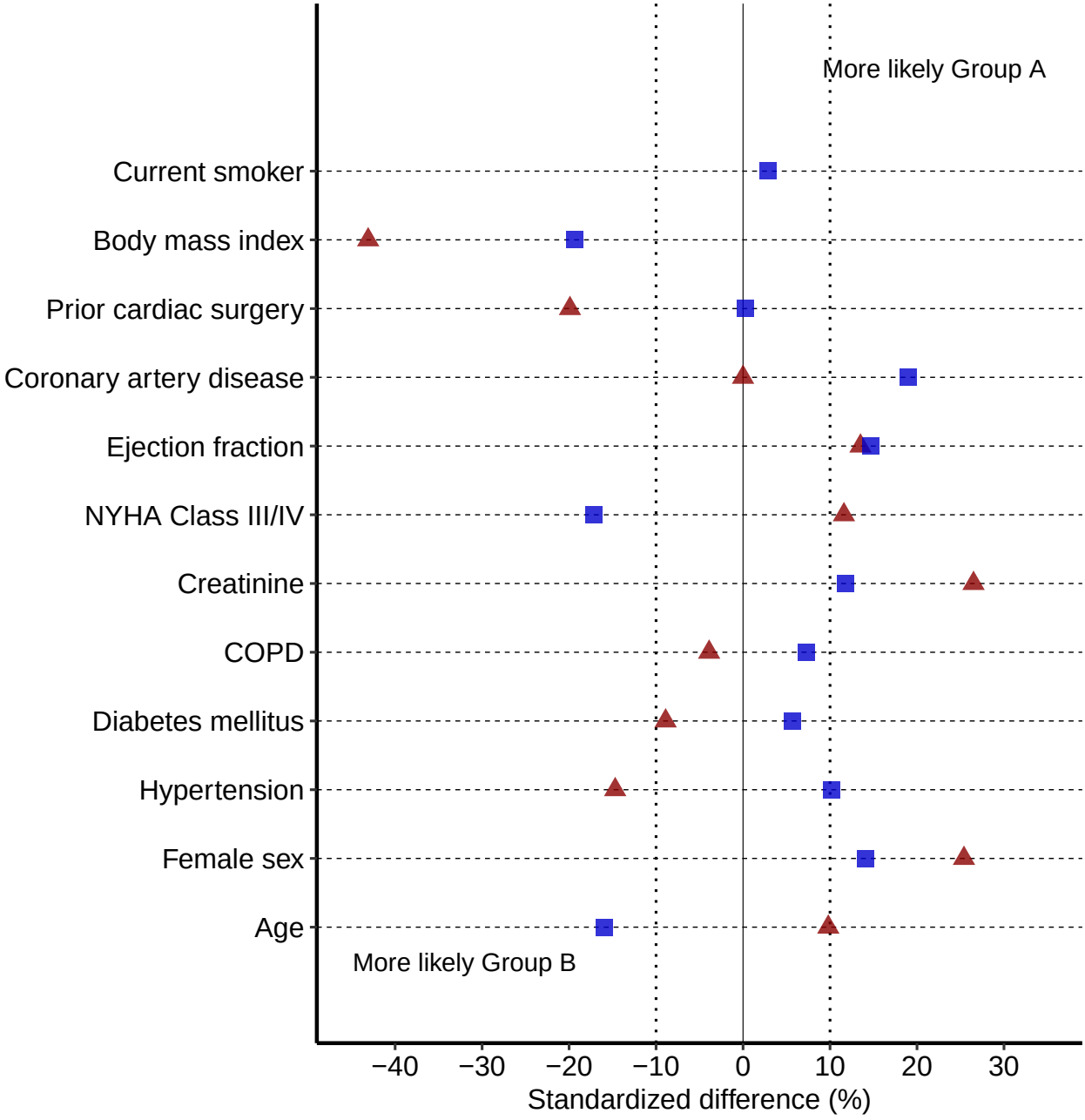
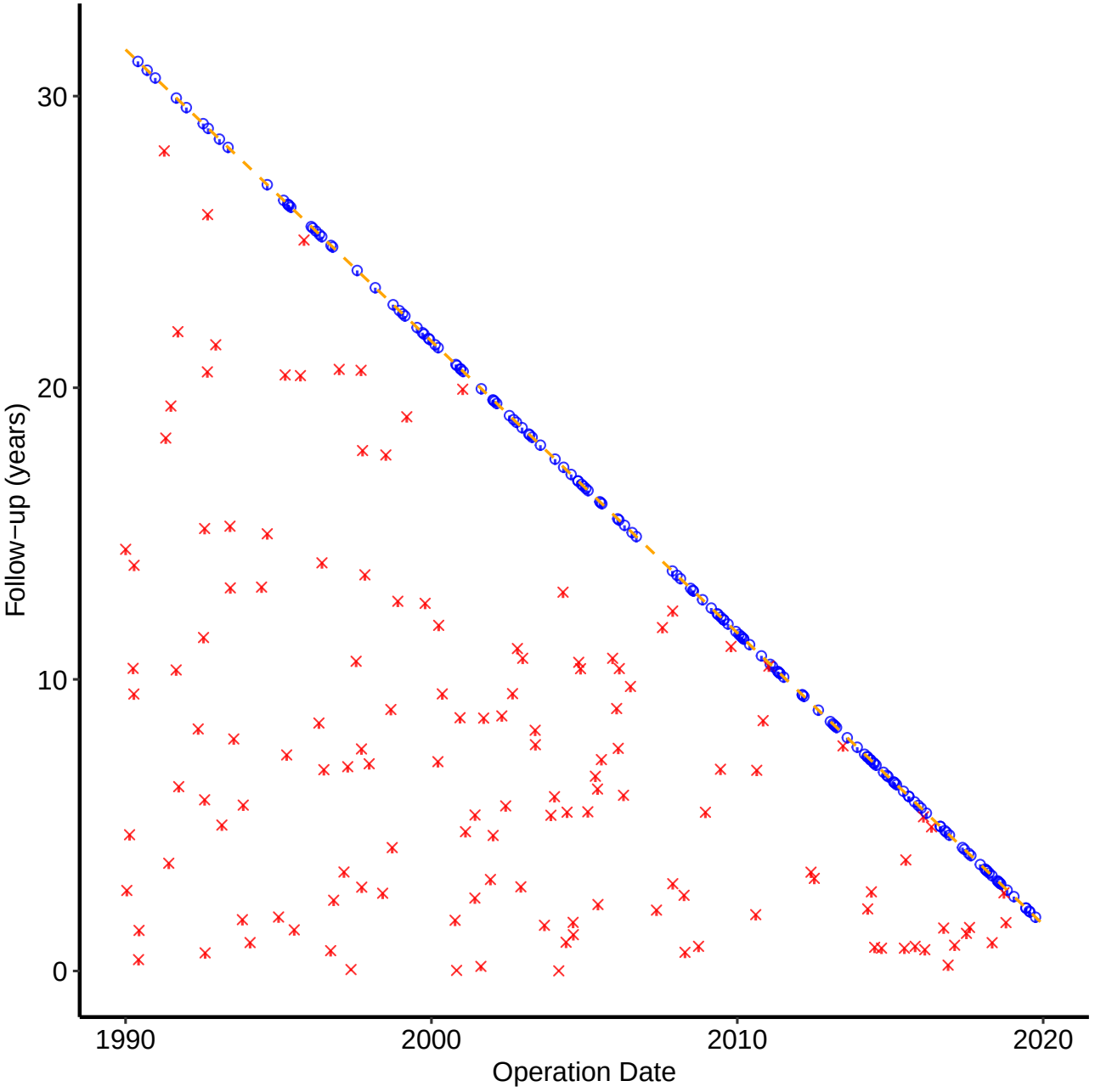
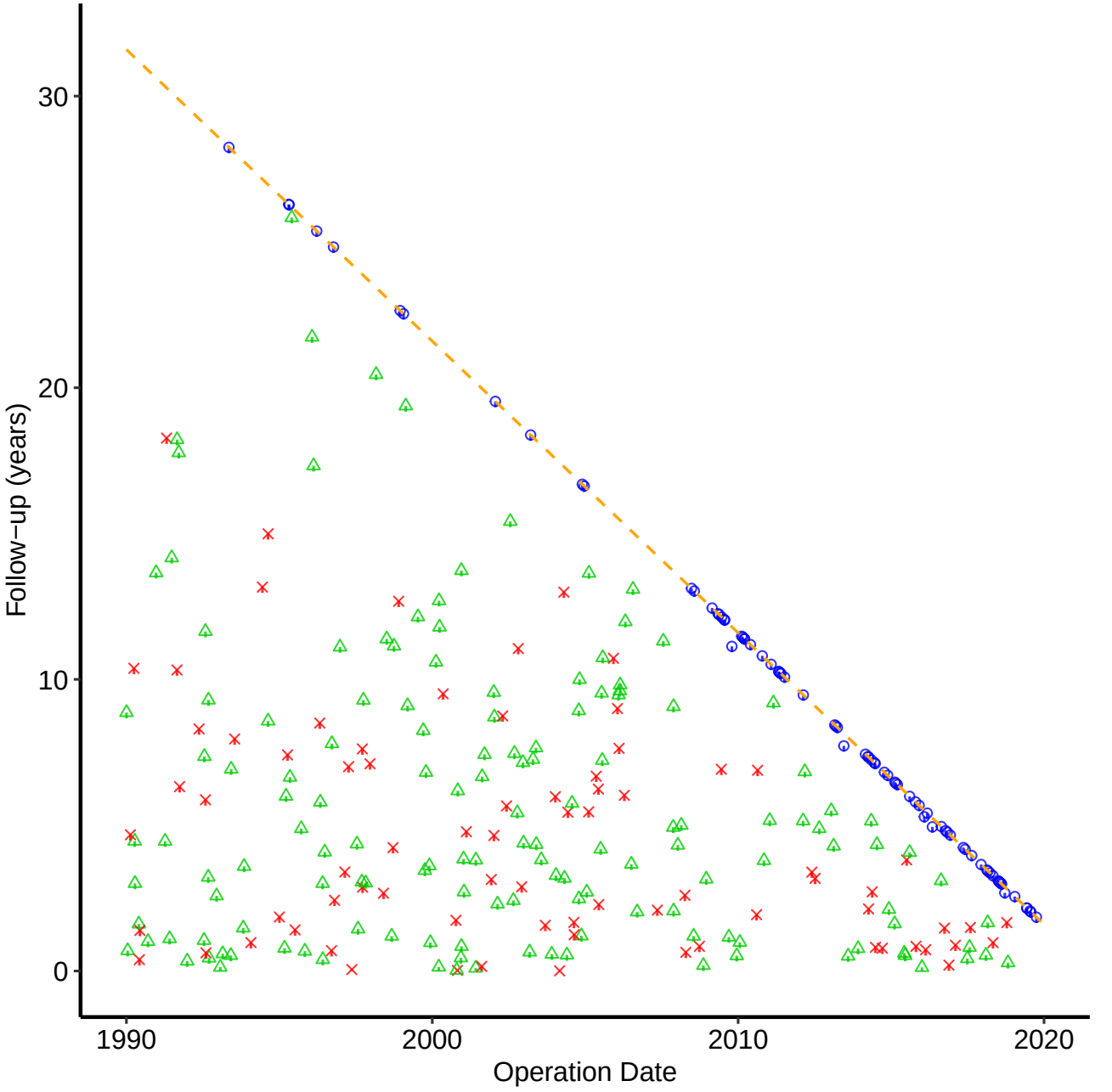
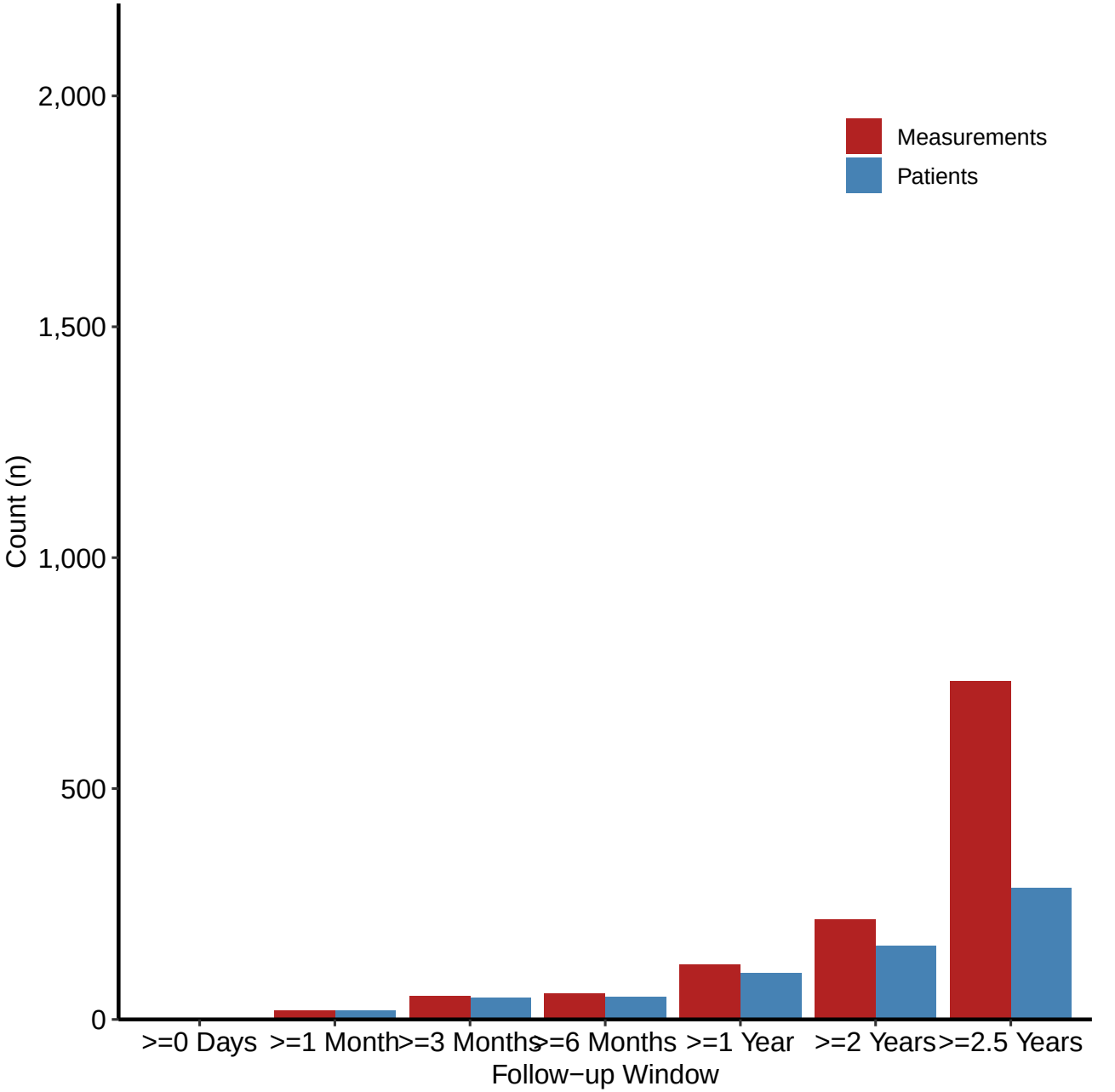

Current smoker
Body mass index
Prior cardiac surgery
Coronary artery disease
Ejection fraction
NYHA Class III/IV
Creatinine
COPD
Diabetes mellitus
Hypertension
Female sex
Age
cb_index
−80
−40
0
std_diff

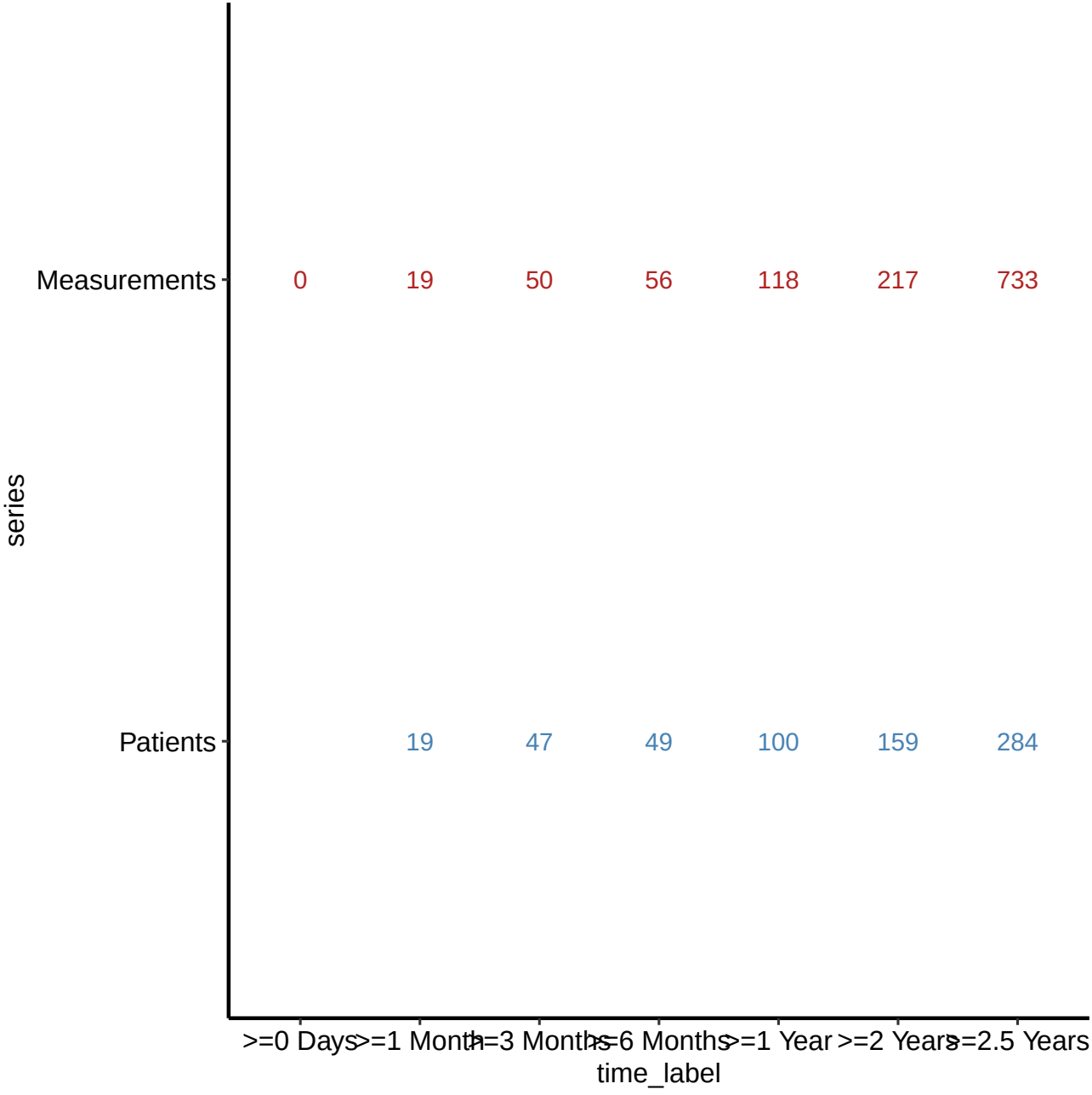

After match
Before match
Current smoker
Body mass index
Prior cardiac surgery
Coronary artery disease
Ejection fraction
NYHA Class III/IV
Creatinine
COPD
Diabetes mellitus
Hypertension
Female sex
Age
−40
−30
−20
−10
0
10
20
30
Standardized difference: Group A vs Group B (%)

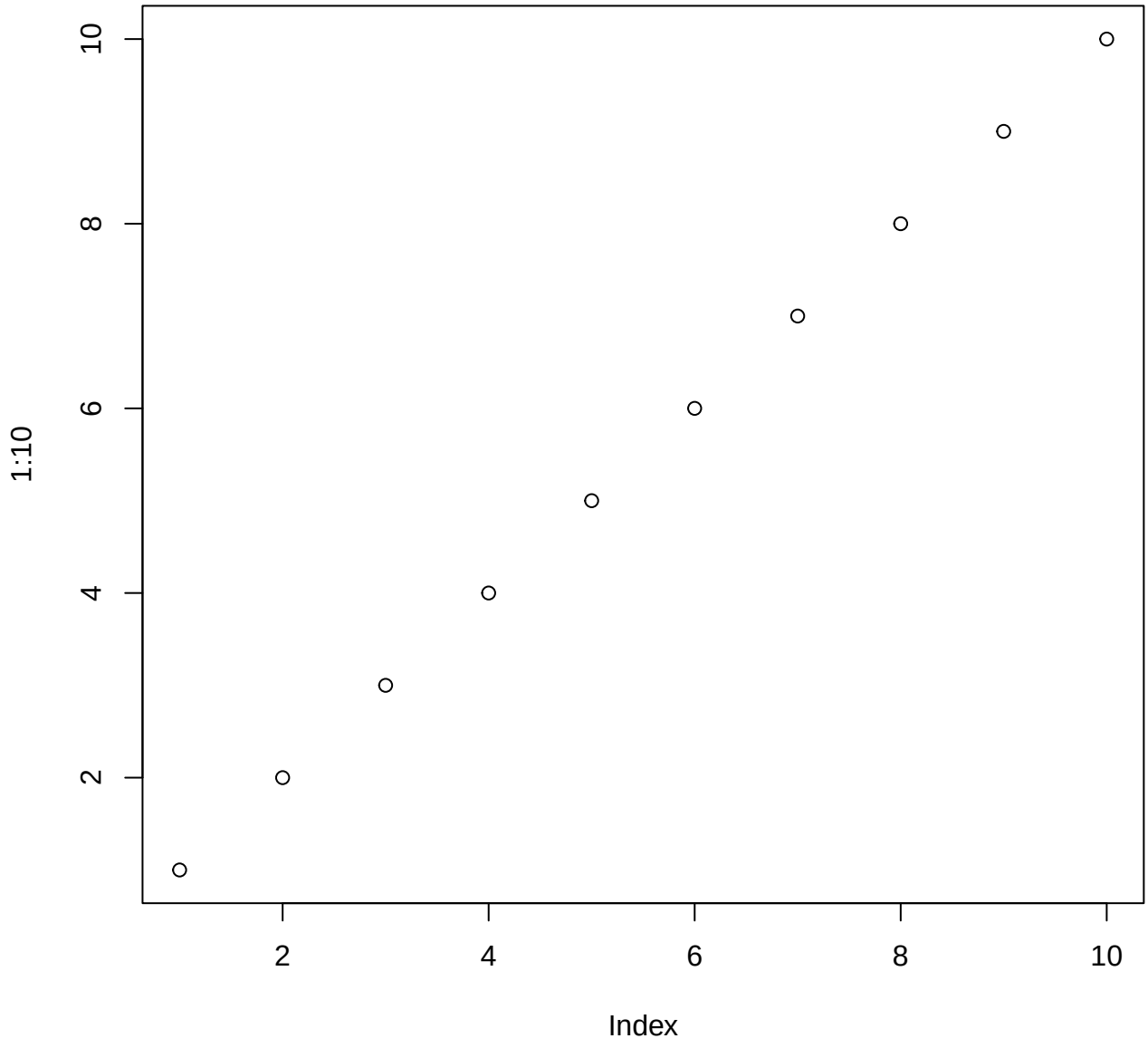

More likely Group A
Current smoker
Body mass index
Prior cardiac surgery
Coronary artery disease
Ejection fraction
NYHA Class III/IV
Creatinine
COPD
Diabetes mellitus
Hypertension
Female sex
Age
More likely Group B
−40
−30
−20
−10
0
10
20
30
Standardized difference (%)


30
20
10
0
Follow−up (years)
1990
2000
2010
2020
Operation Date

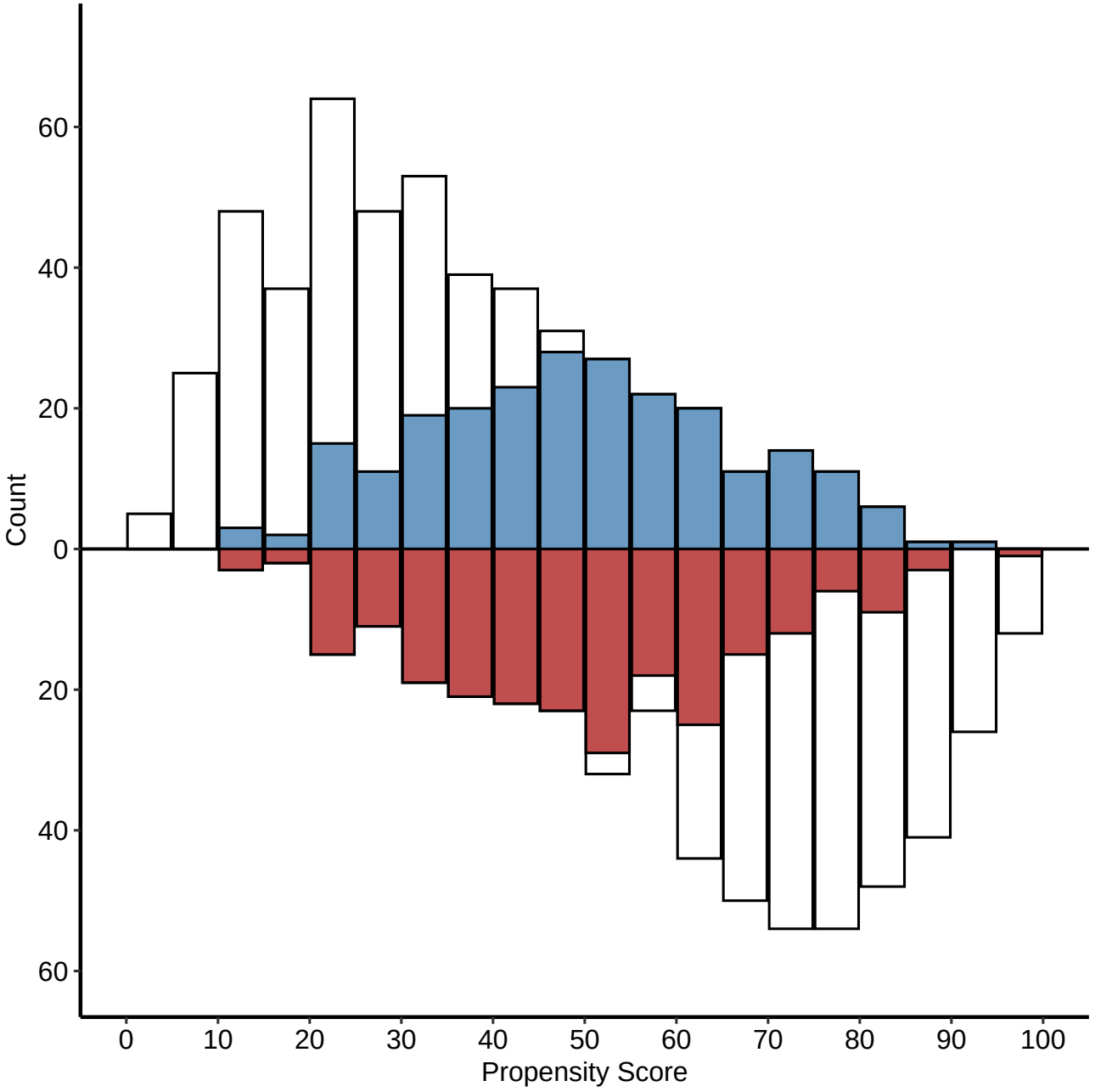

30
20
10
0
Follow−up (years)
1990
2000
2010
2020
Operation Date

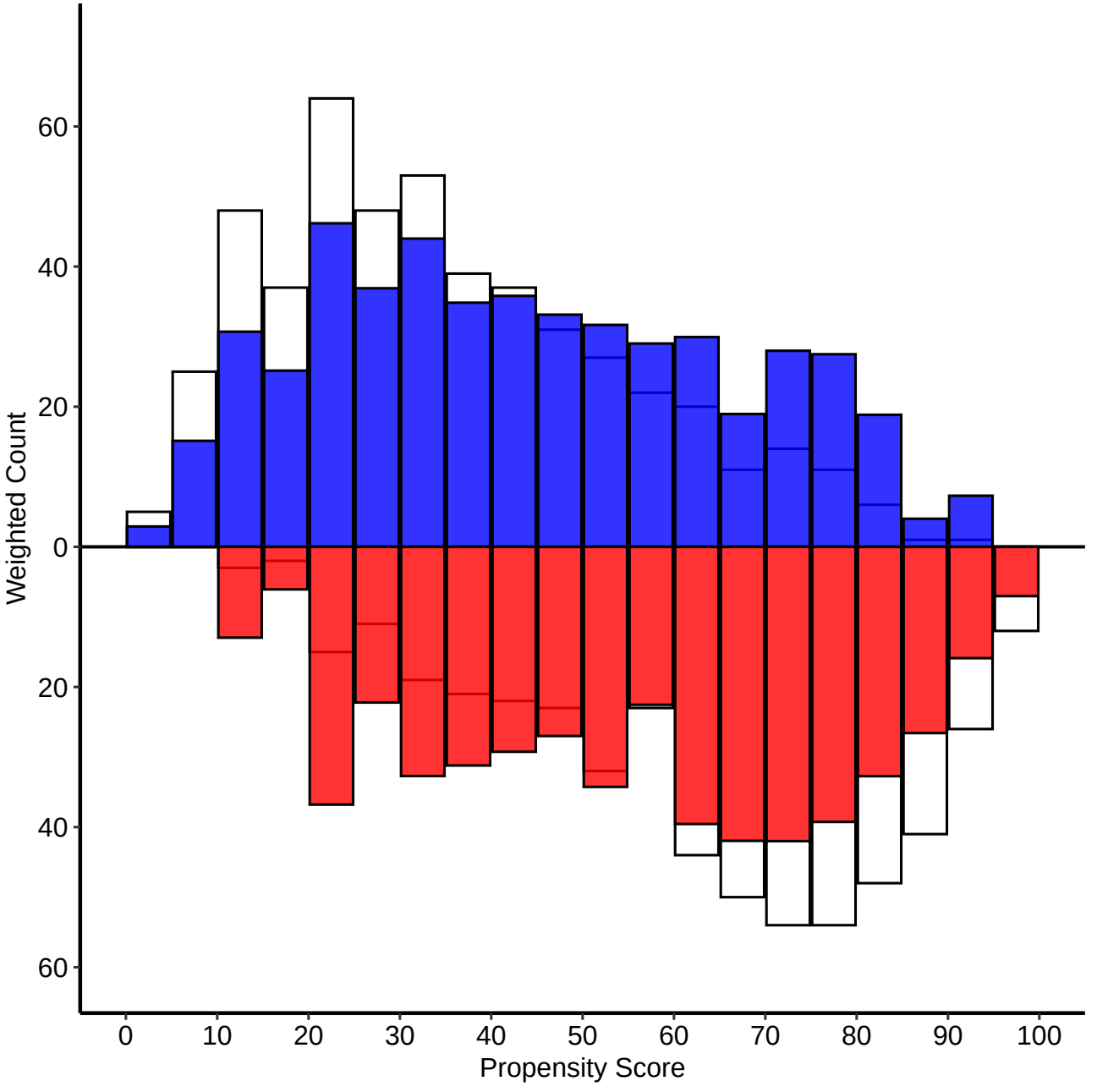

Measurements
Patients
Count (n)
2,000
1,500
1,000
500
0
>=0 Days
>=1 Month
>=3 Months
>=6 Months
>=1 Year
>=2 Years
>=2.5 Years
Follow−up Window

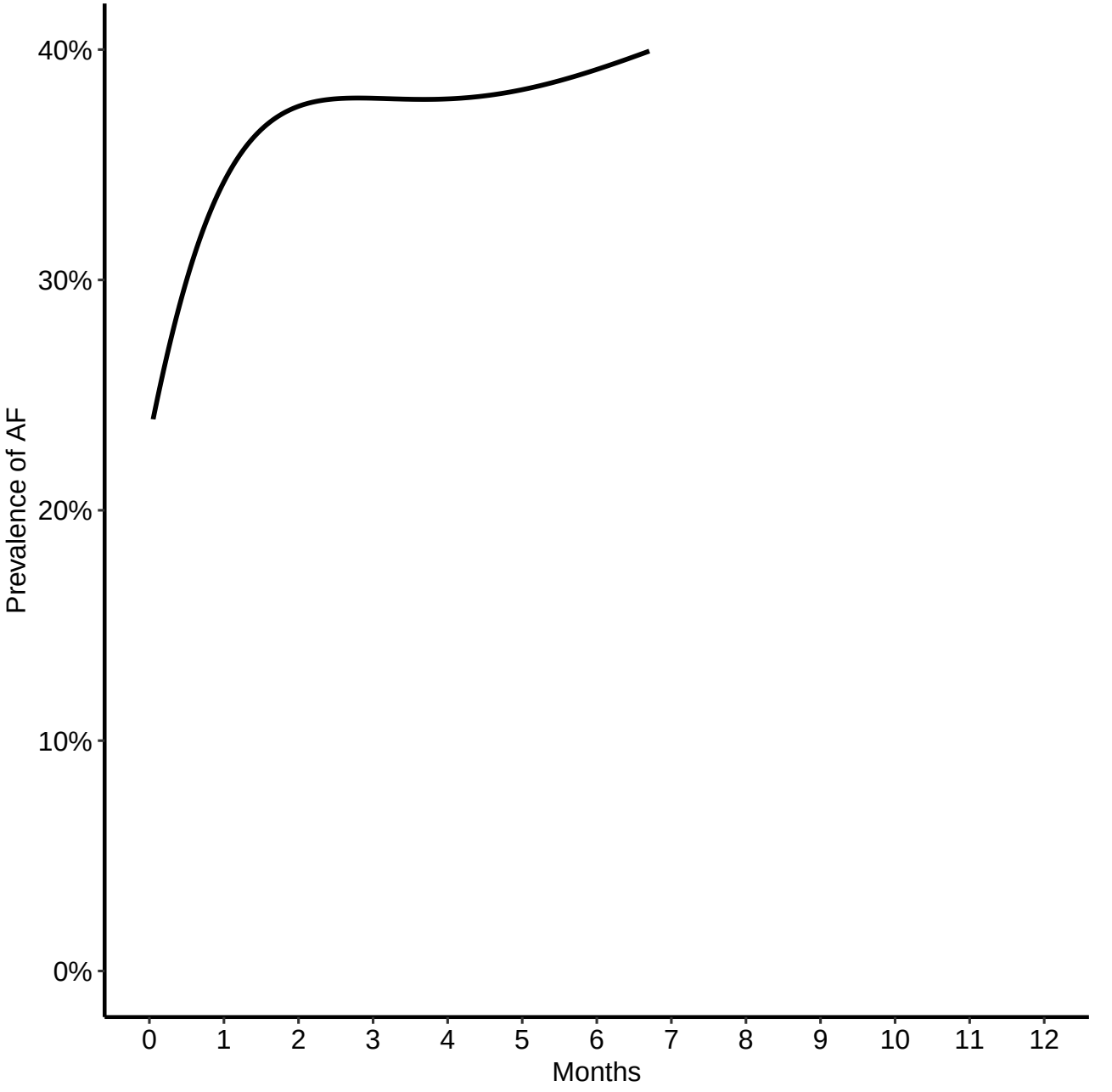

Measurements
0
19
50
56
118
217
733
Patients
19
47
49
100
159
284
series
>=0 Days
>=1 Month
>=3 Months
>=6 Months
>=1 Year
>=2 Years
>=2.5 Years
time_label

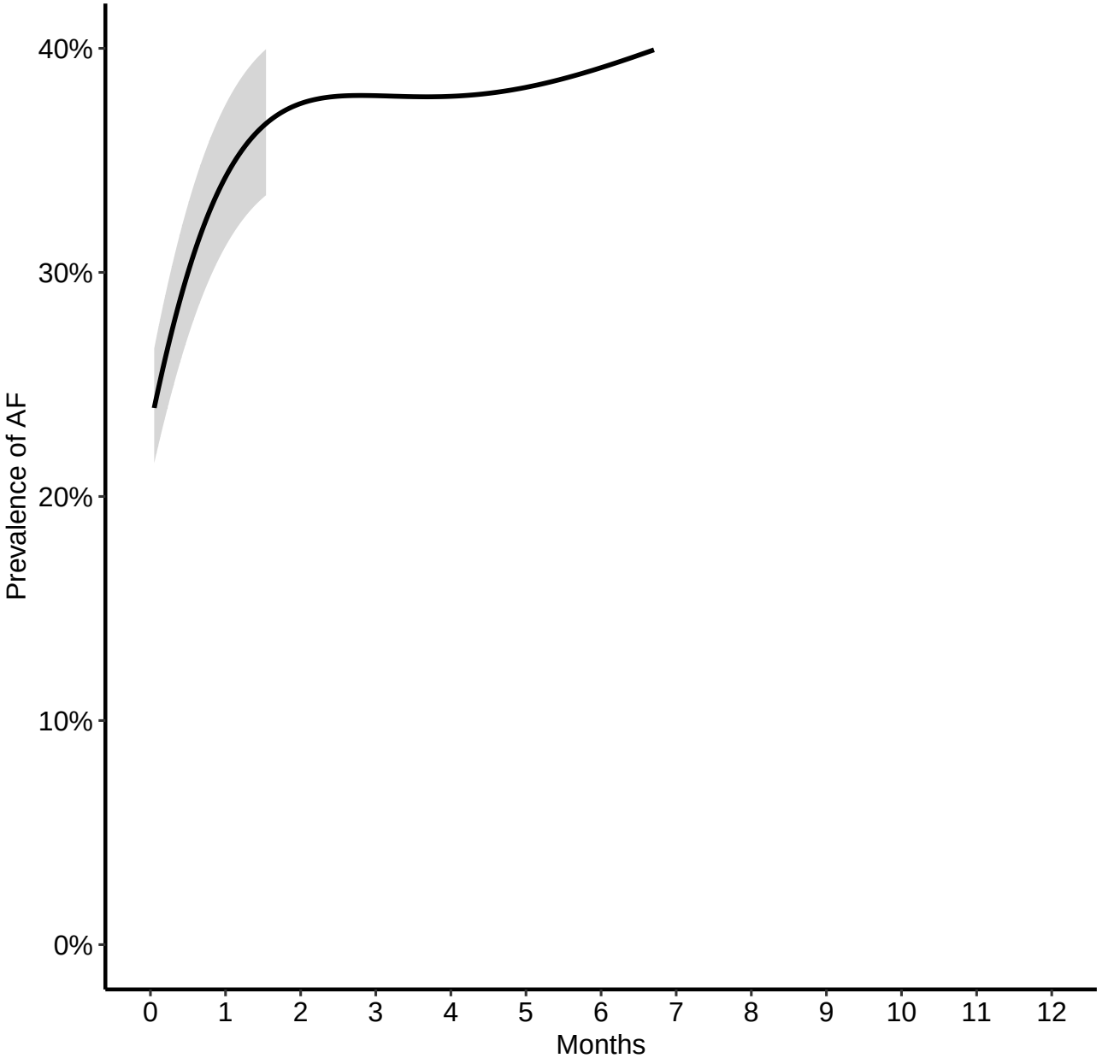


/Users/ehrlinj/Documents/GitHub/hviPlotR 2026−03−18 13:12:18.834215

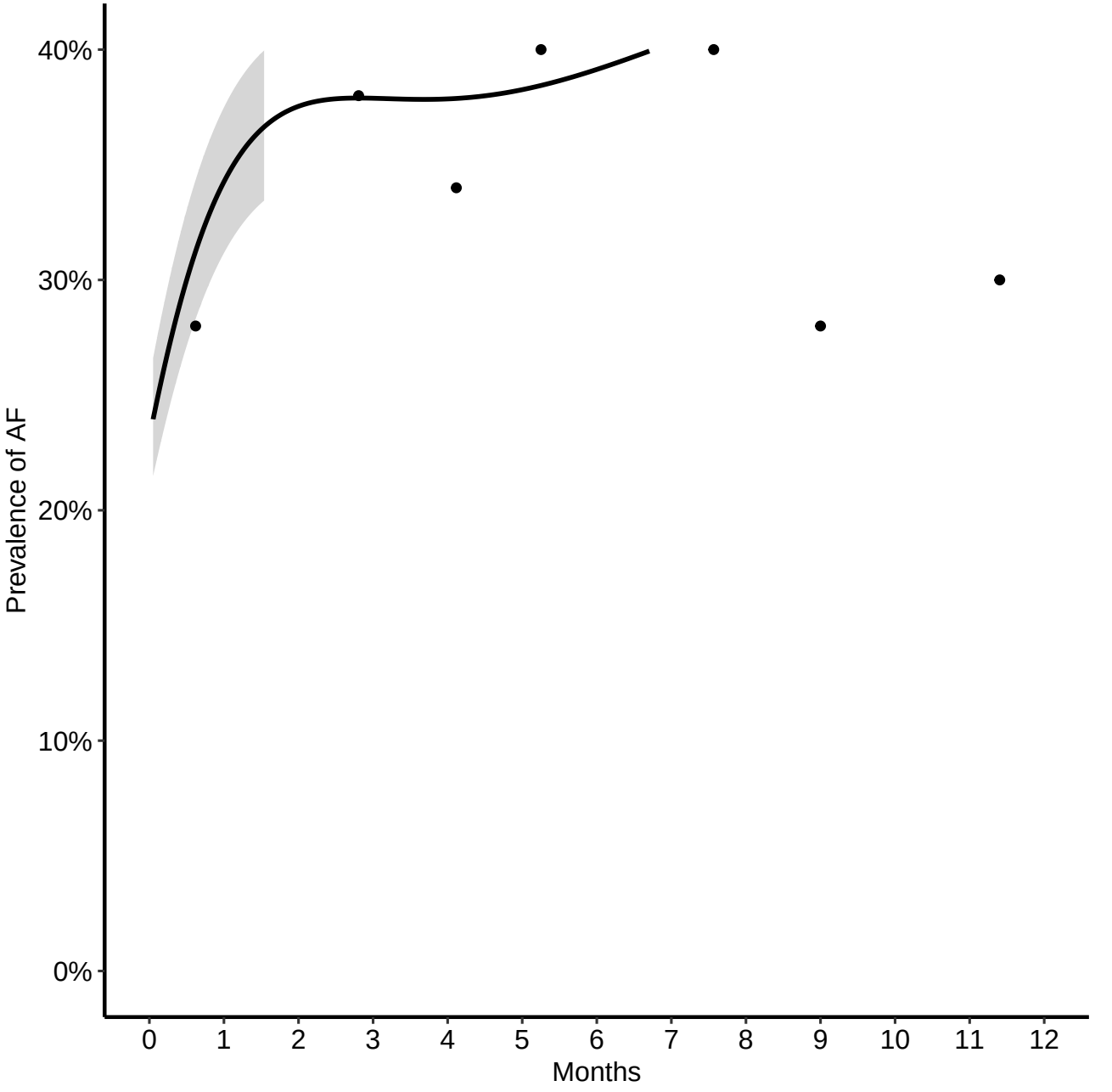

Count
60
40
20
0
20
40
60
0
10
20
30
40
50
60
70
80
90
100
Propensity Score

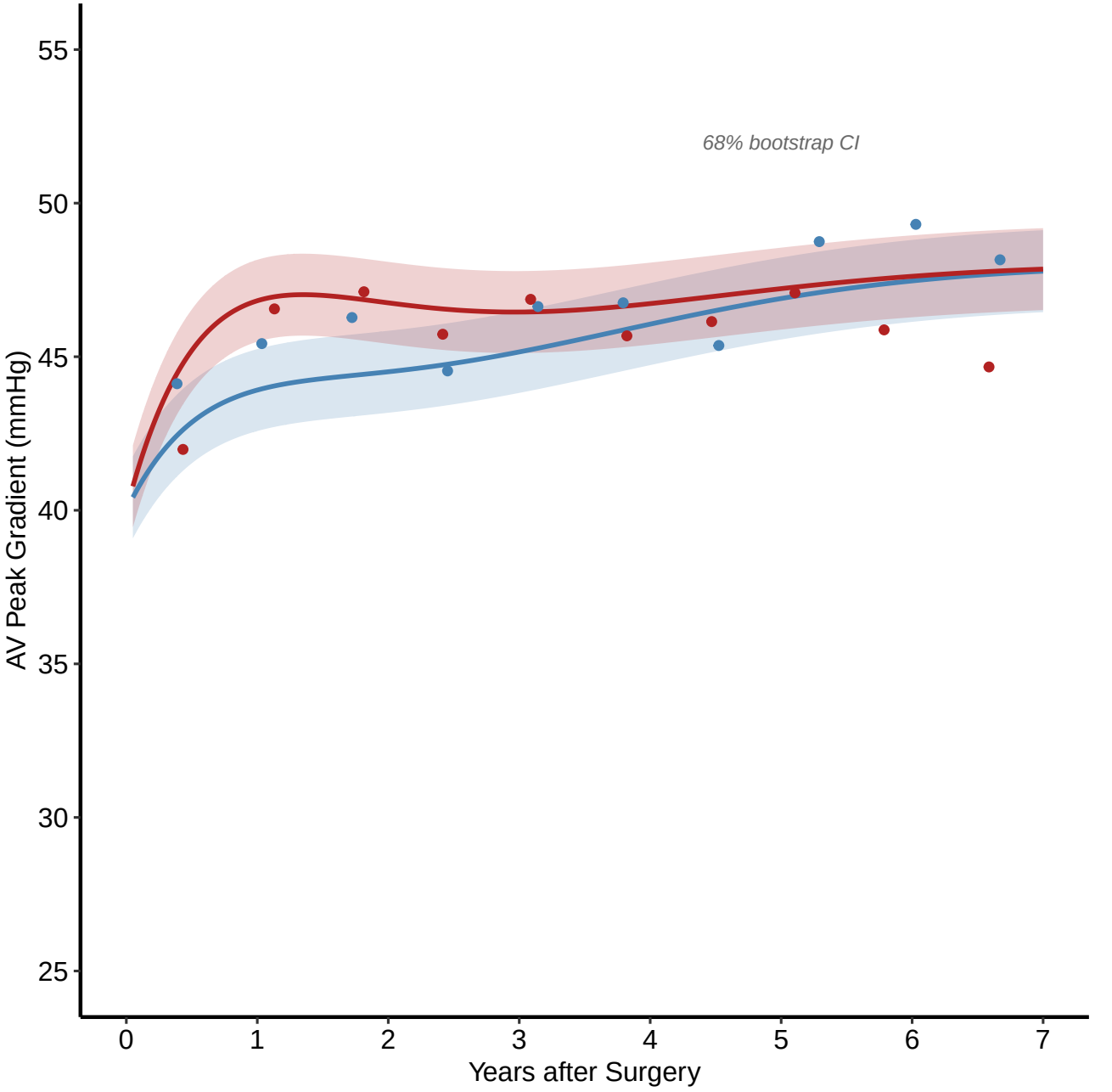

Weighted Count
60
40
20
0
20
40
60
0
10
20
30
40
50
60
70
80
90
100
Propensity Score

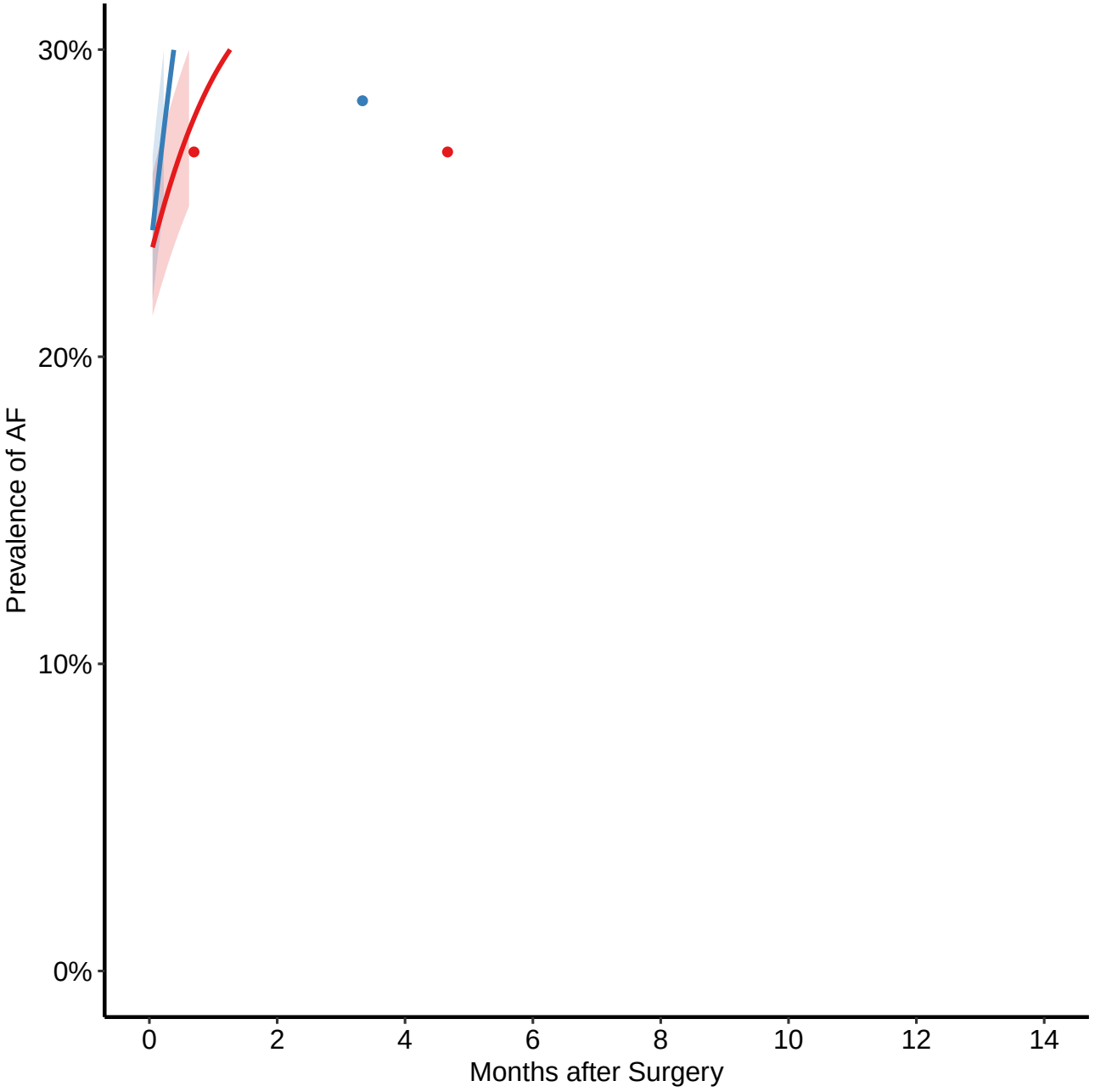

40%
30%
20%
10%
0%
Prevalence of AF
0
1
2
3
4
5
6
7
8
9
10
11
12
Months

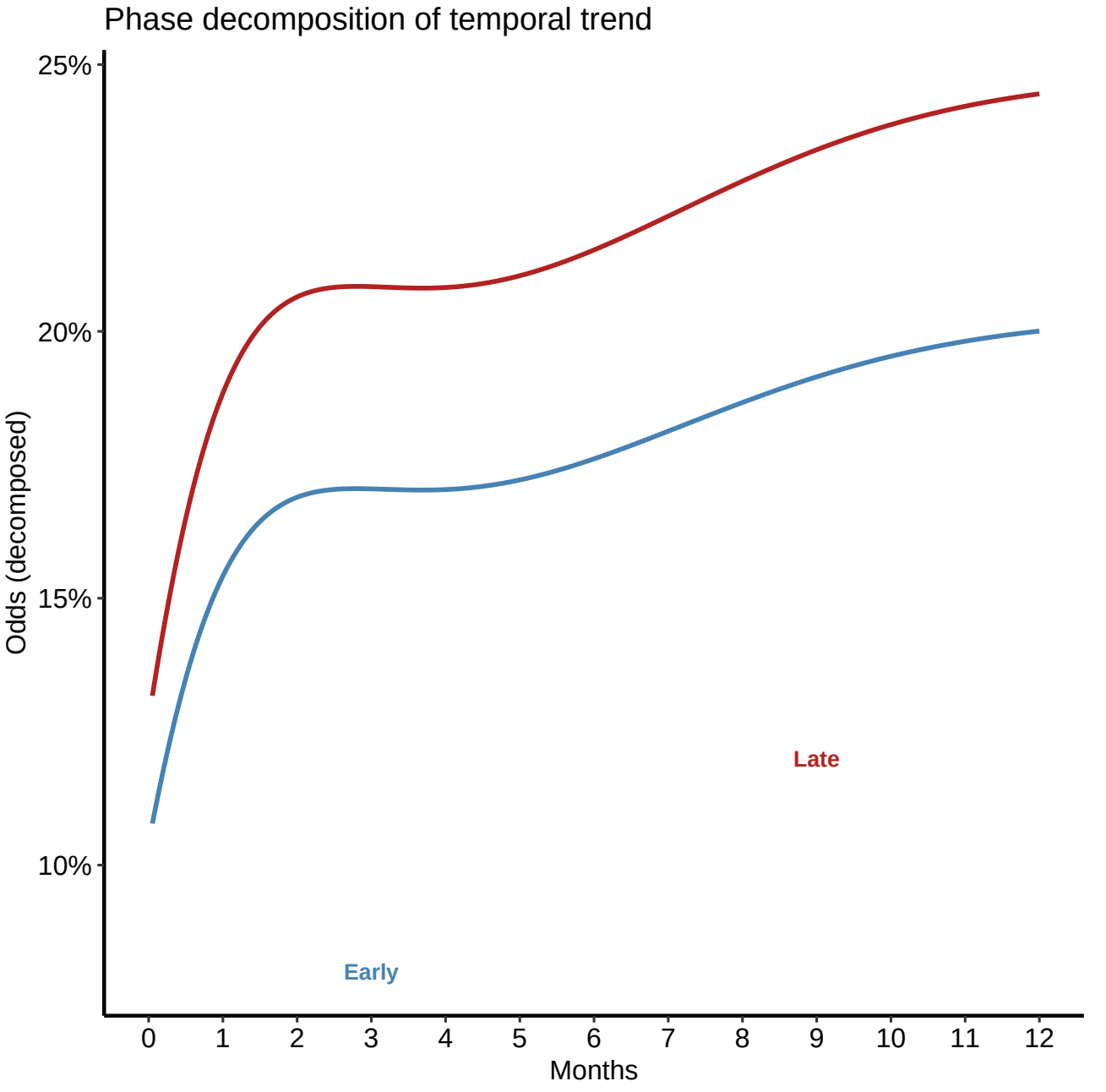


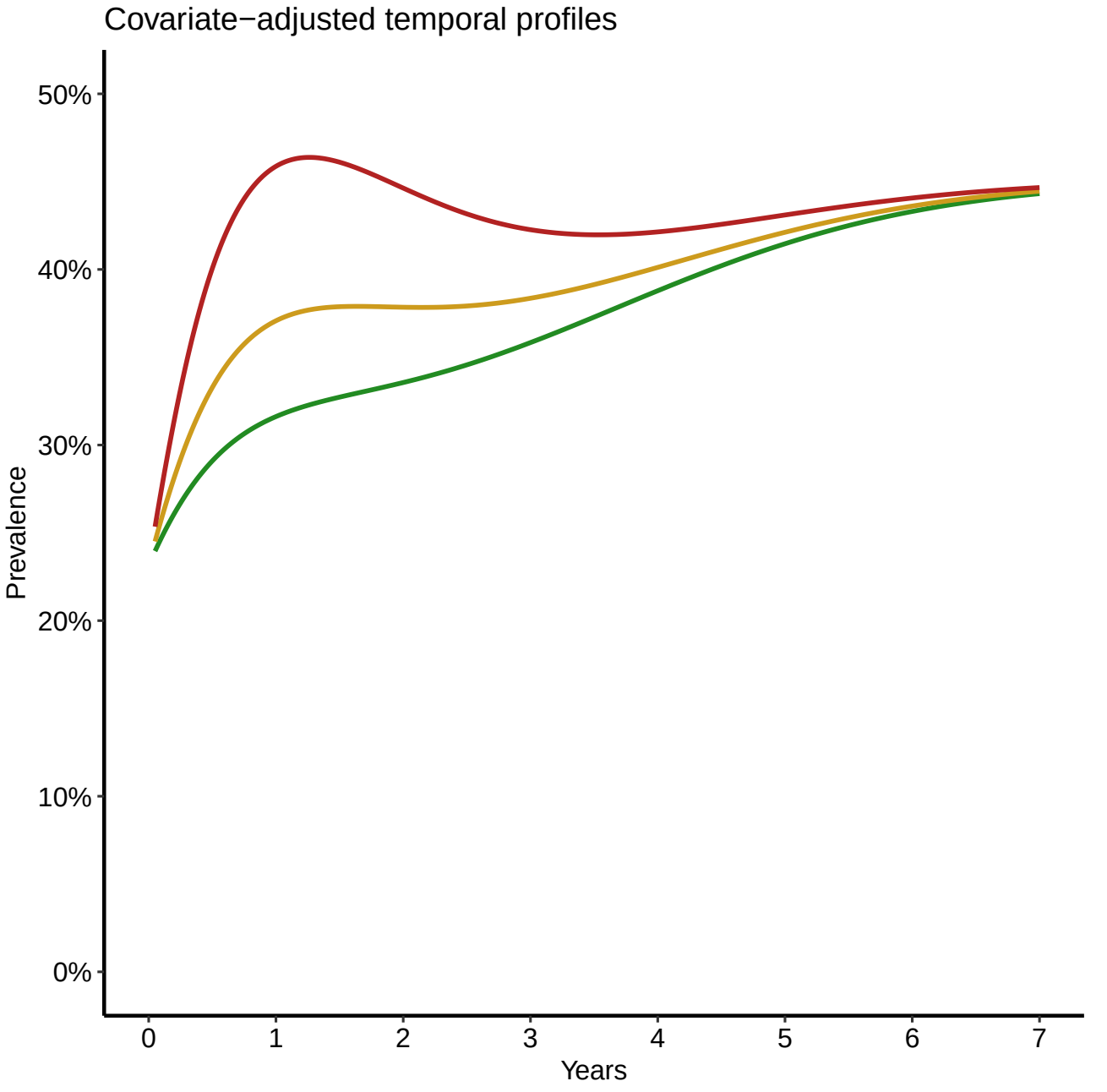
Shaded region: 68% bootstrap confidence interval

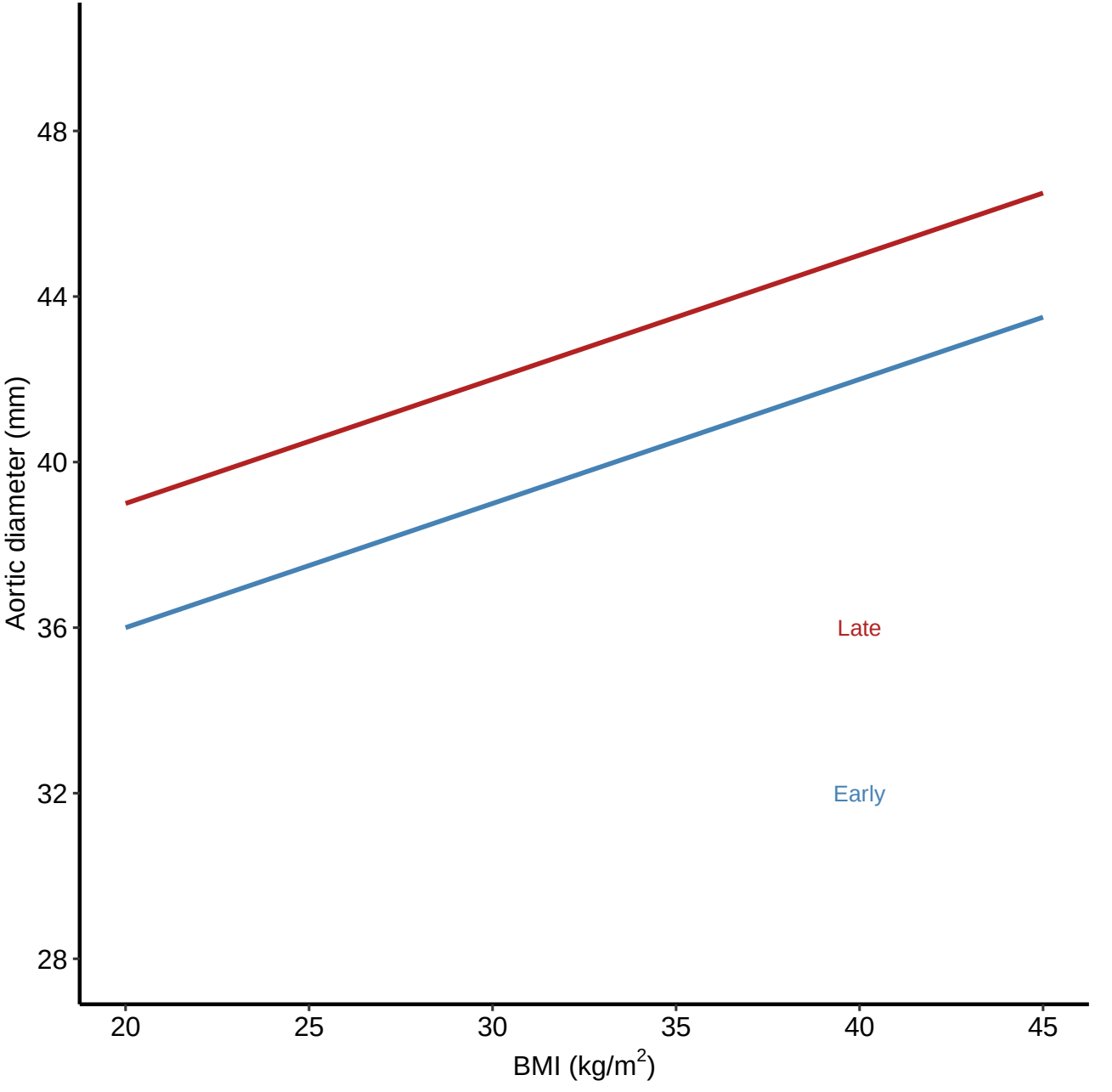

40%
30%
20%
10%
0%
Prevalence of AF
0
1
2
3
4
5
6
7
8
9
10
11
12
Months

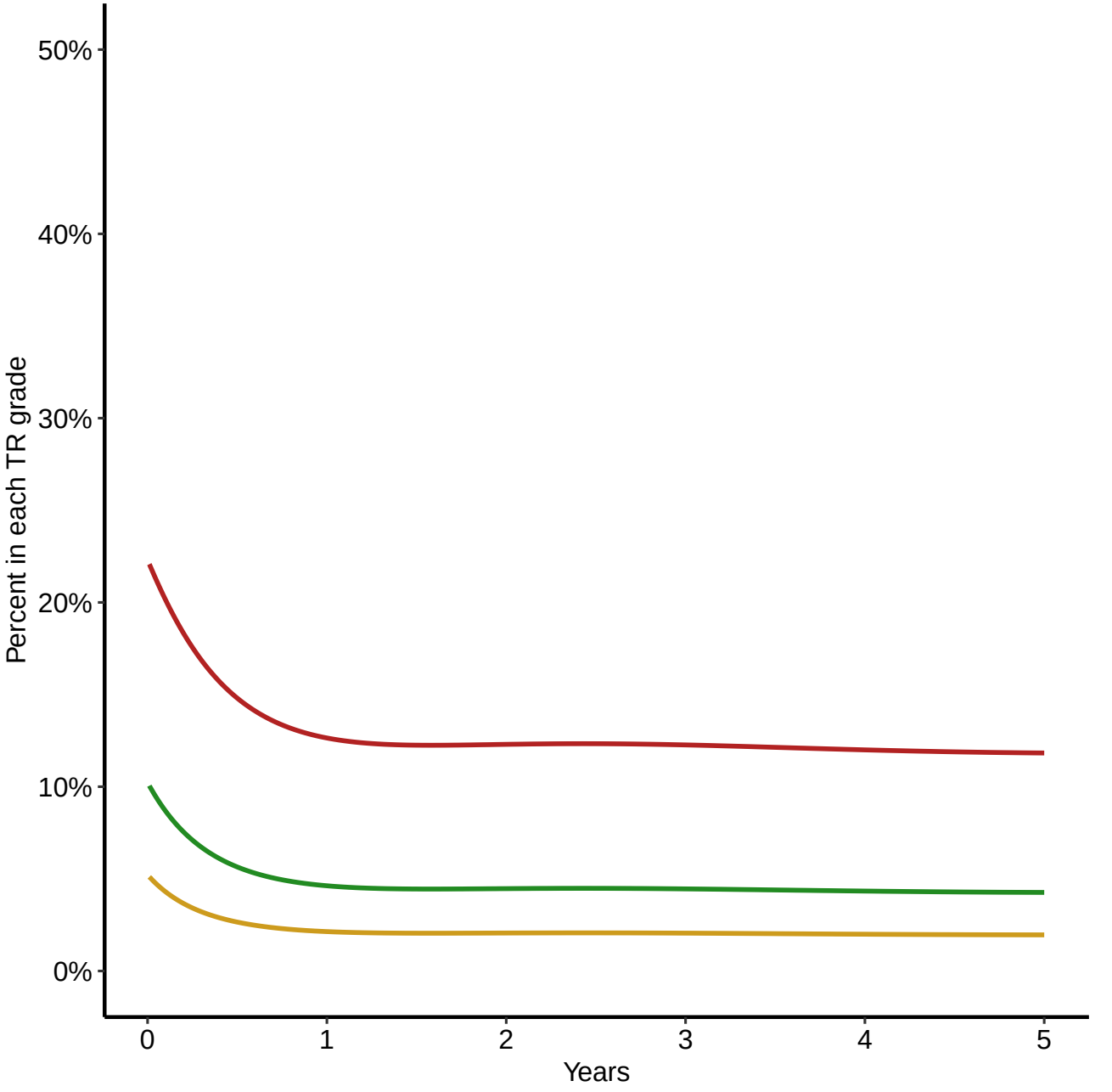

68% bootstrap CI
AV Peak Gradient (mmHg)
55
50
45
40
35
30
25
0
1
2
3
4
5
6
7
Years after Surgery

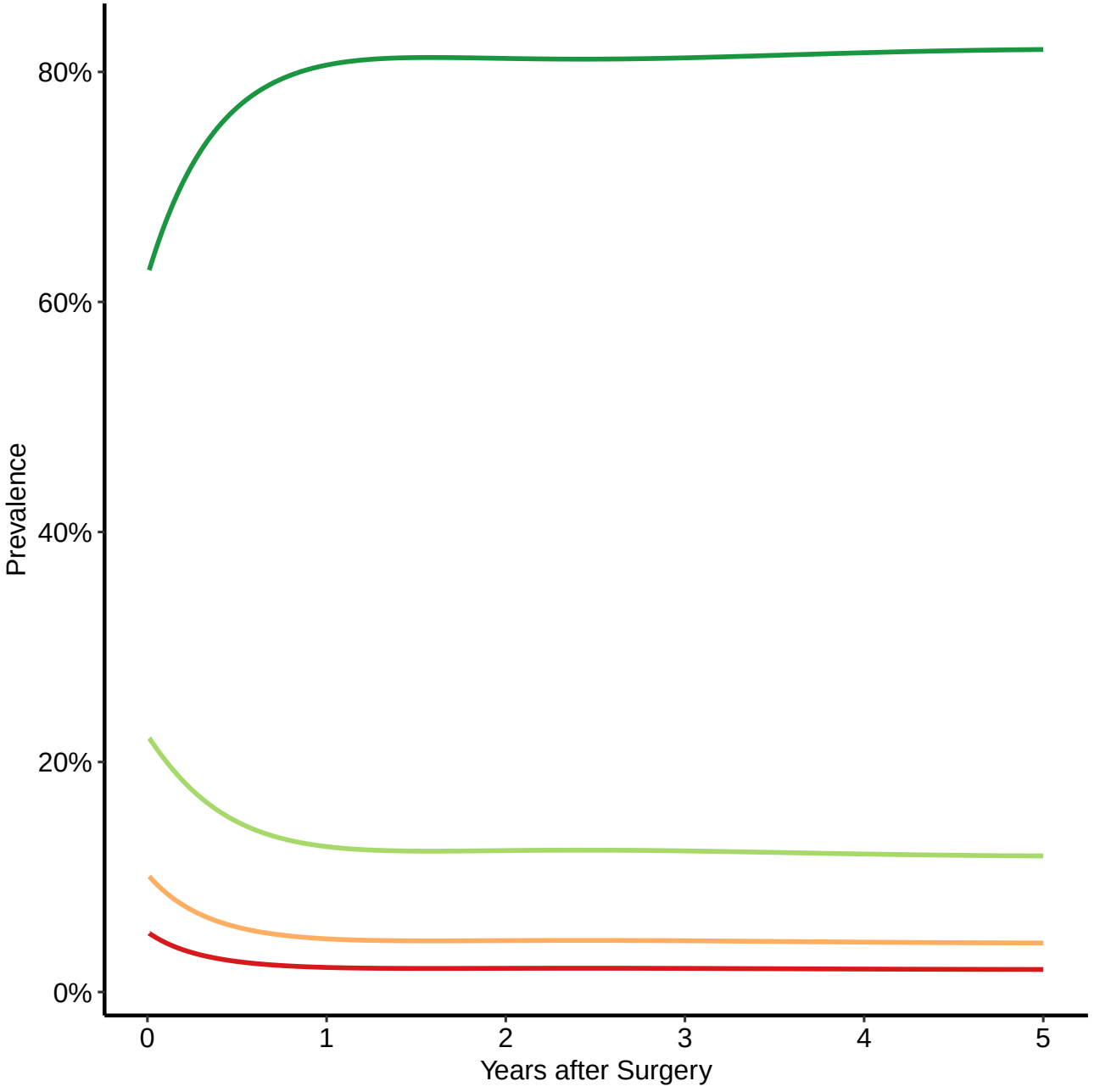

30%
20%
10%
0%
Prevalence of AF
0
2
4
6
8
10
12
14
Months after Surgery

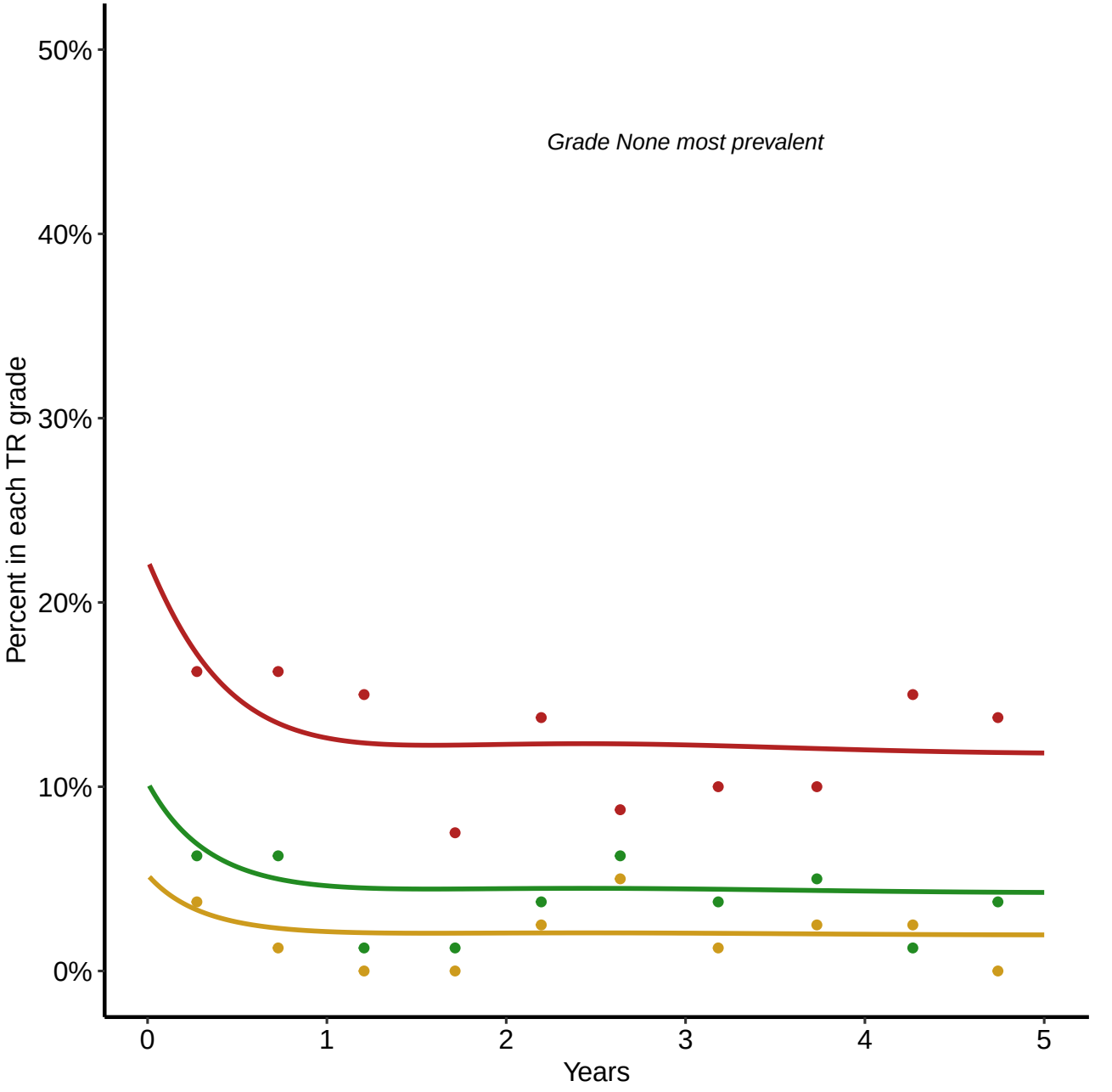

Phase decomposition of temporal trend
25%
20%
15%
10%
Odds (decomposed)
Late
Early
0
1
2
3
4
5
6
7
8
9
10
11
12
Months

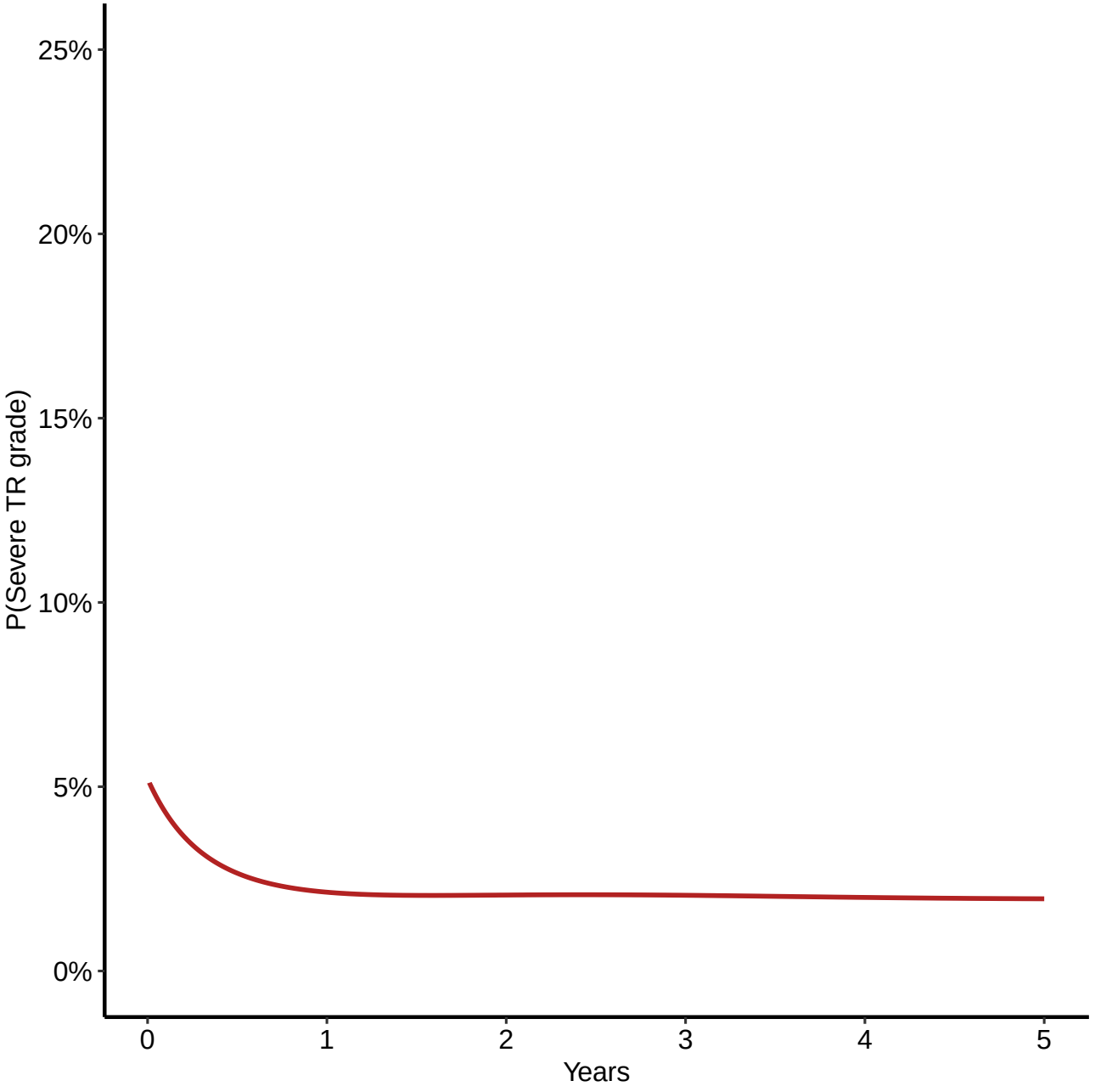

Covariate−adjusted temporal profiles
50%
40%
30%
20%
10%
0%
Prevalence
0
1
2
3
4
5
6
7
Years

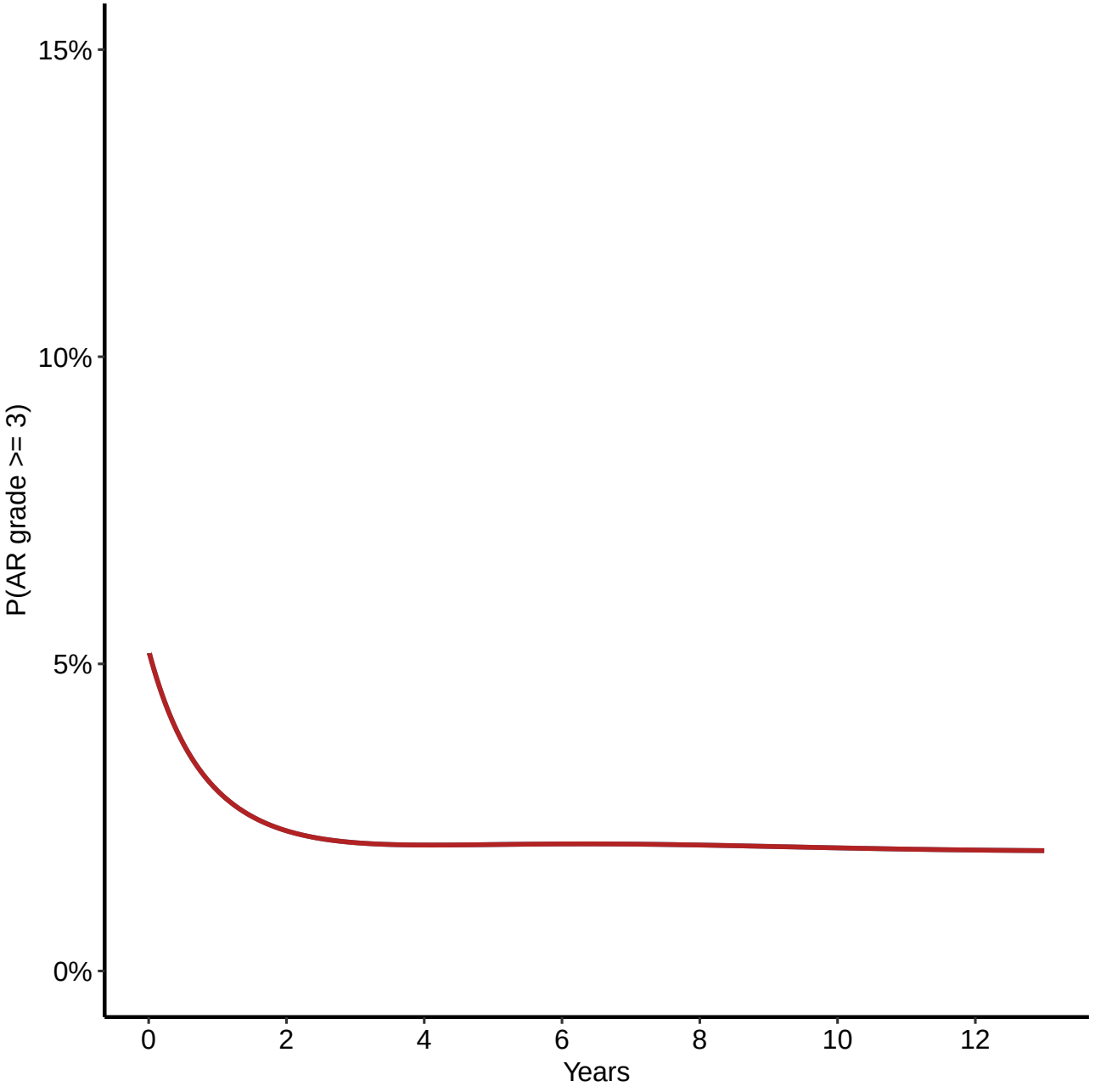

Aortic diameter (mm)
48
44
40
36
32
28
20
25
30
35
40
45
BMI (kg/m2)
Late
Early

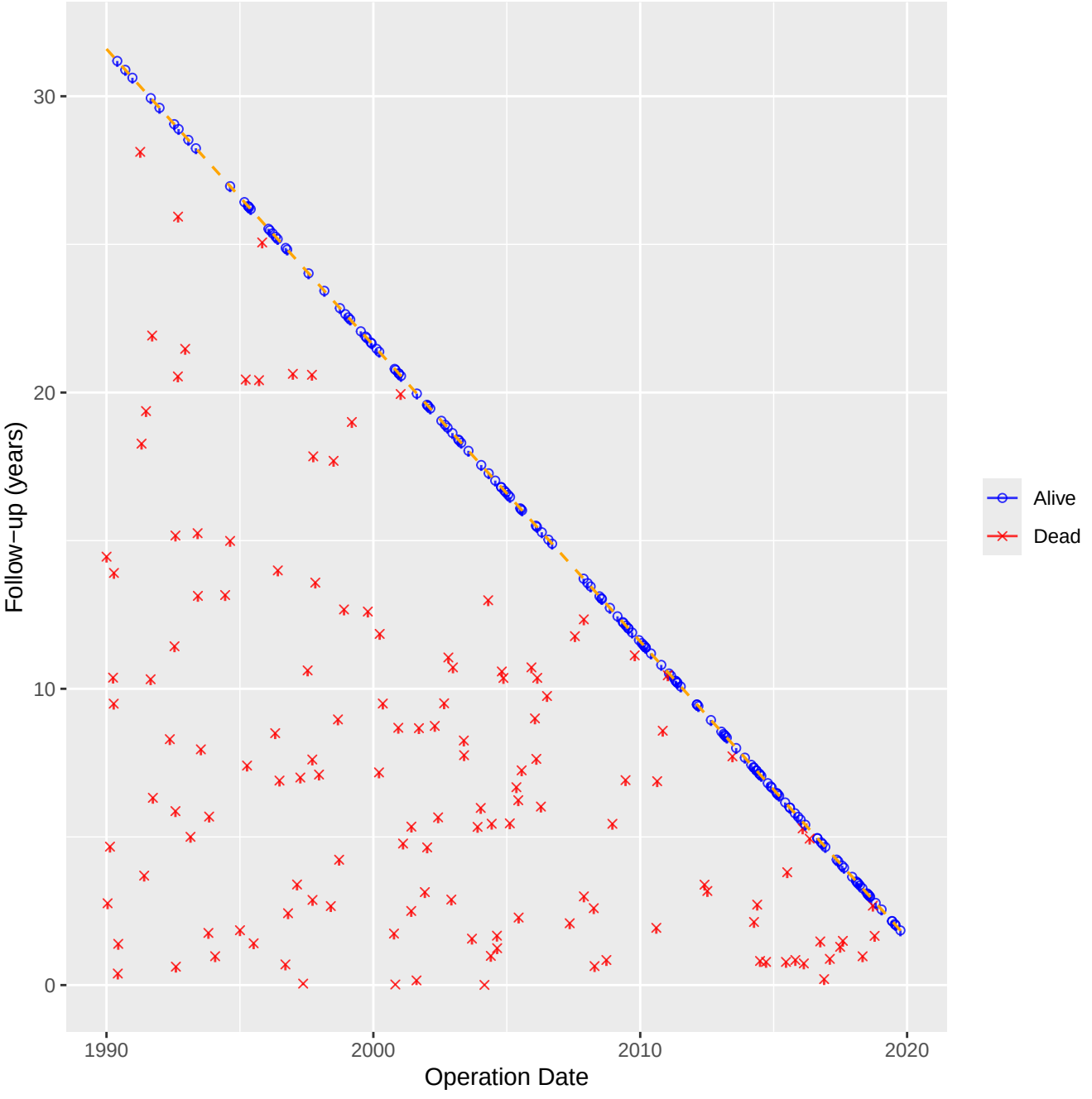

50%
40%
30%
20%
10%
0%
Percent in each TR grade
0
1
2
3
4
5
Years

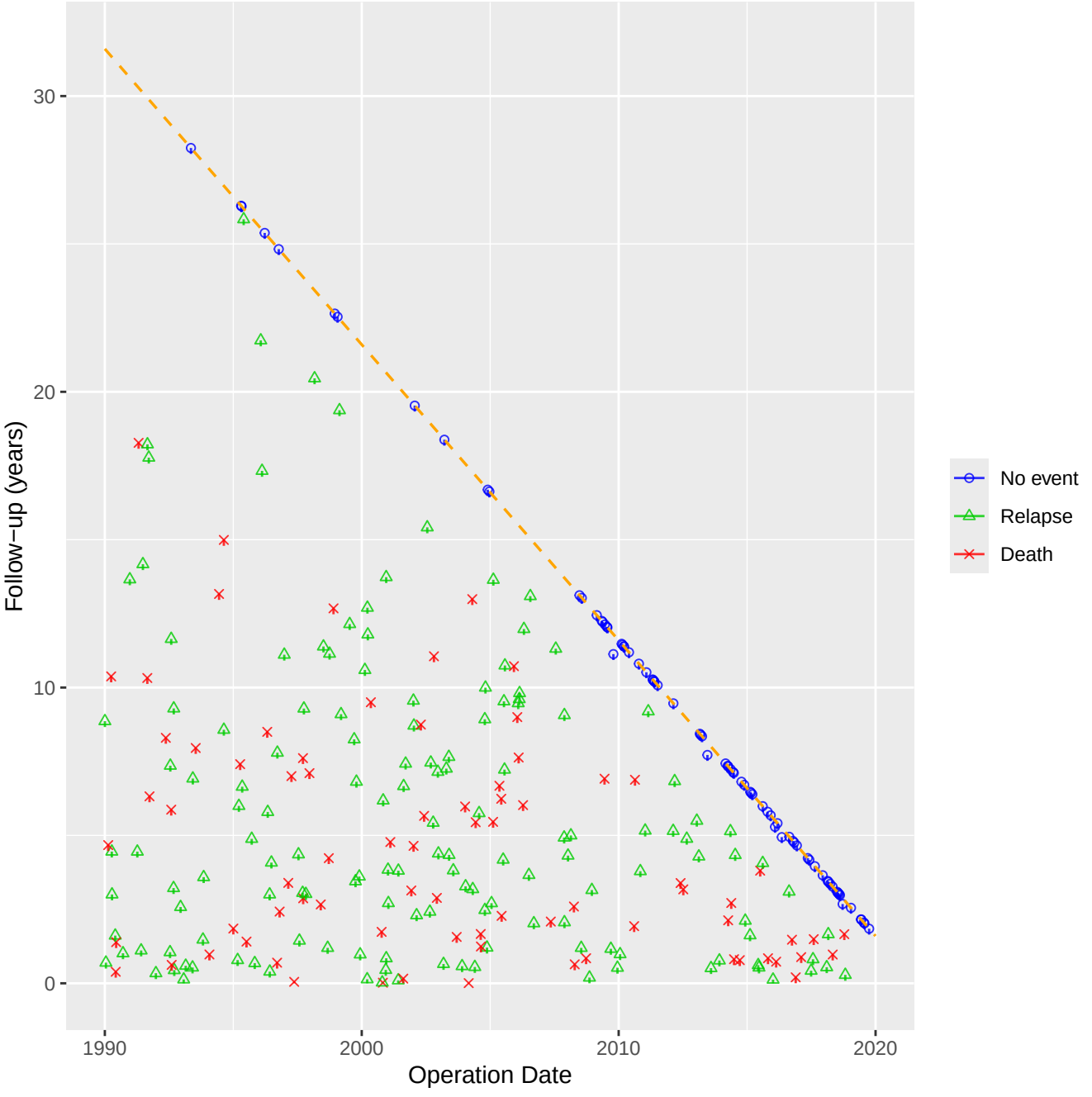

80%
60%
40%
20%
0%
Prevalence
0
1
2
3
4
5
Years after Surgery

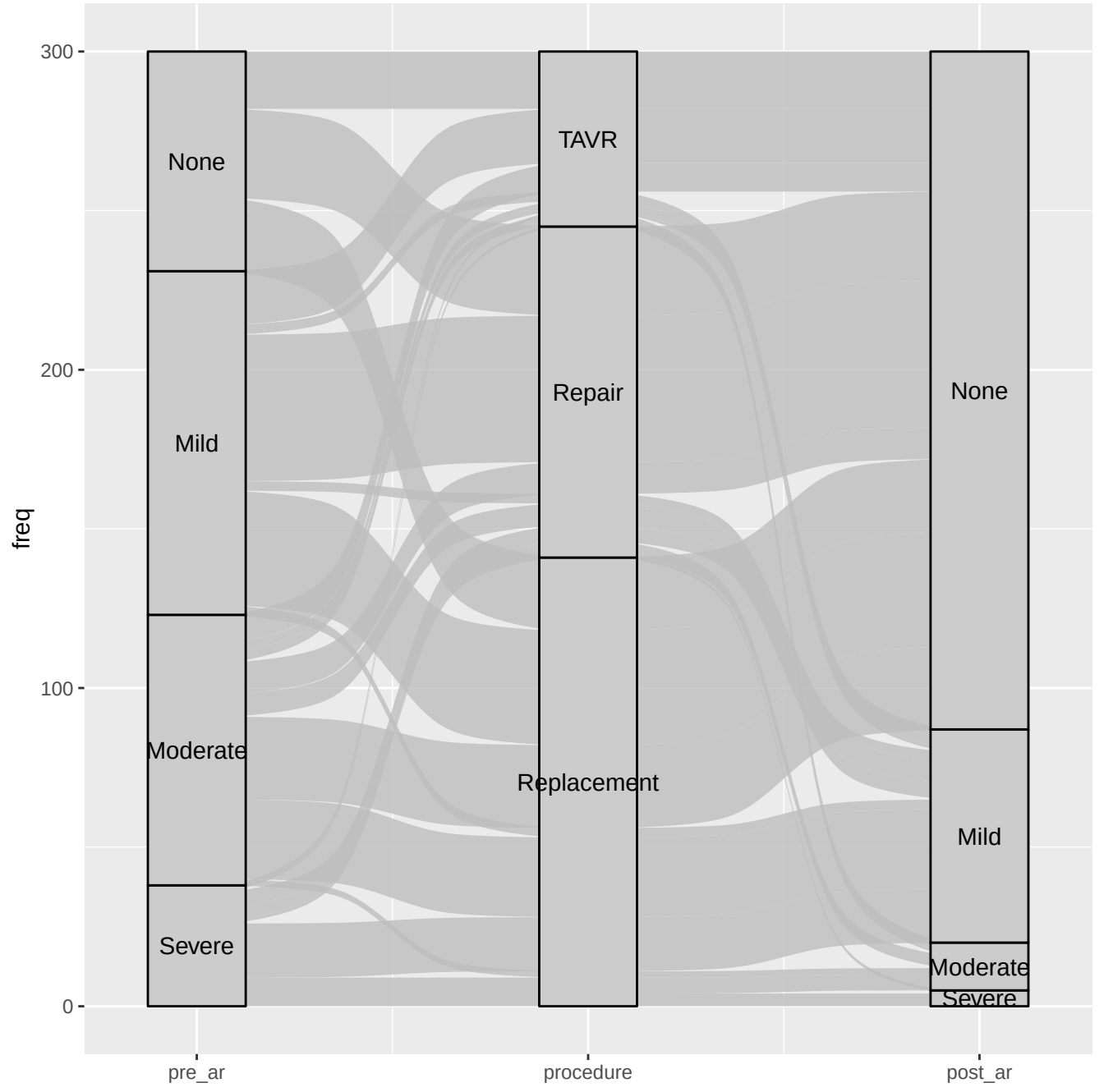

Grade None most prevalent
50%
40%
30%
20%
10%
0%
Percent in each TR grade
0
1
2
3
4
5
Years

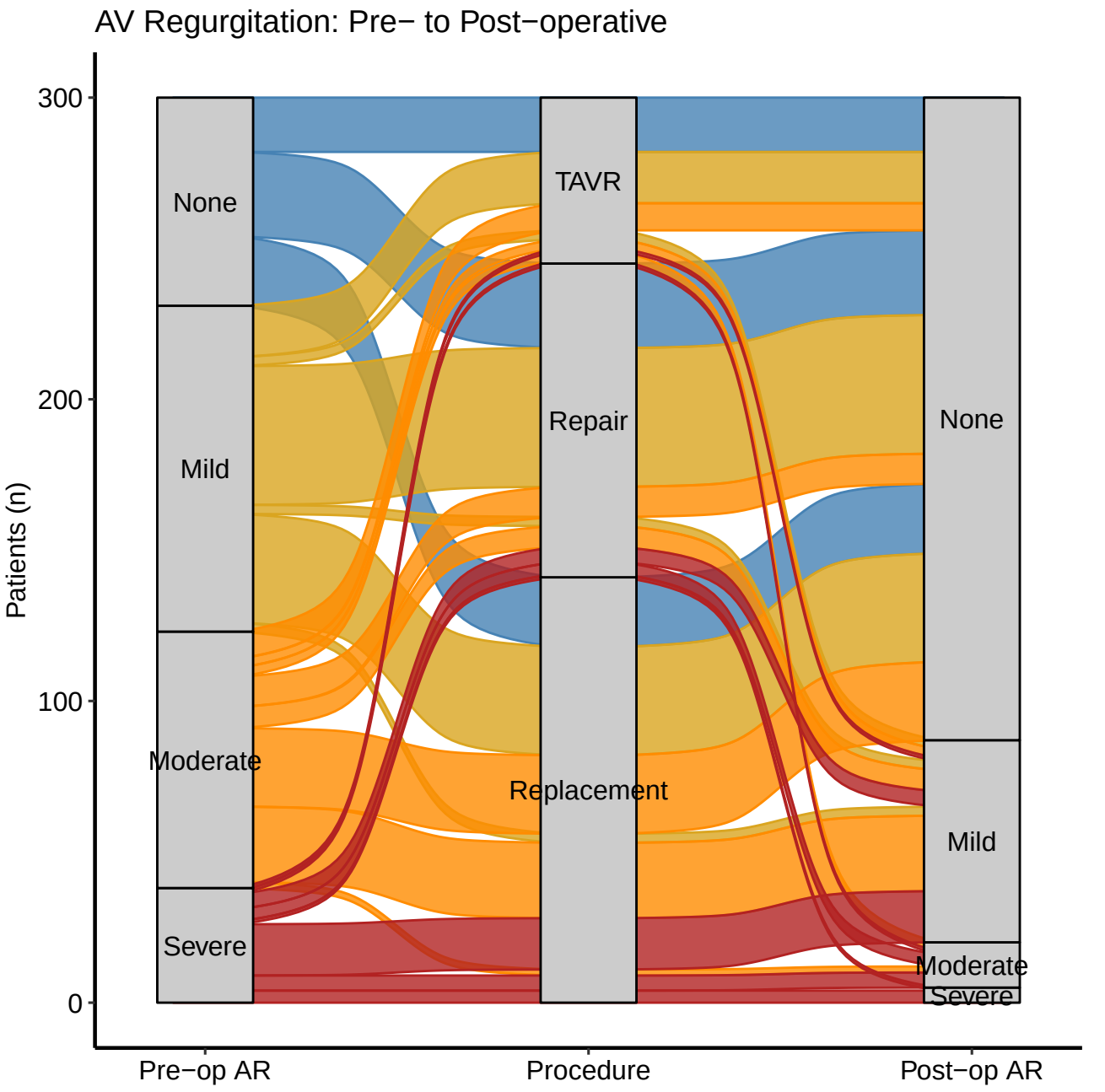

25%
20%
15%
10%
5%
0%
P(Severe TR grade)
0
1
2
3
4
5
Years

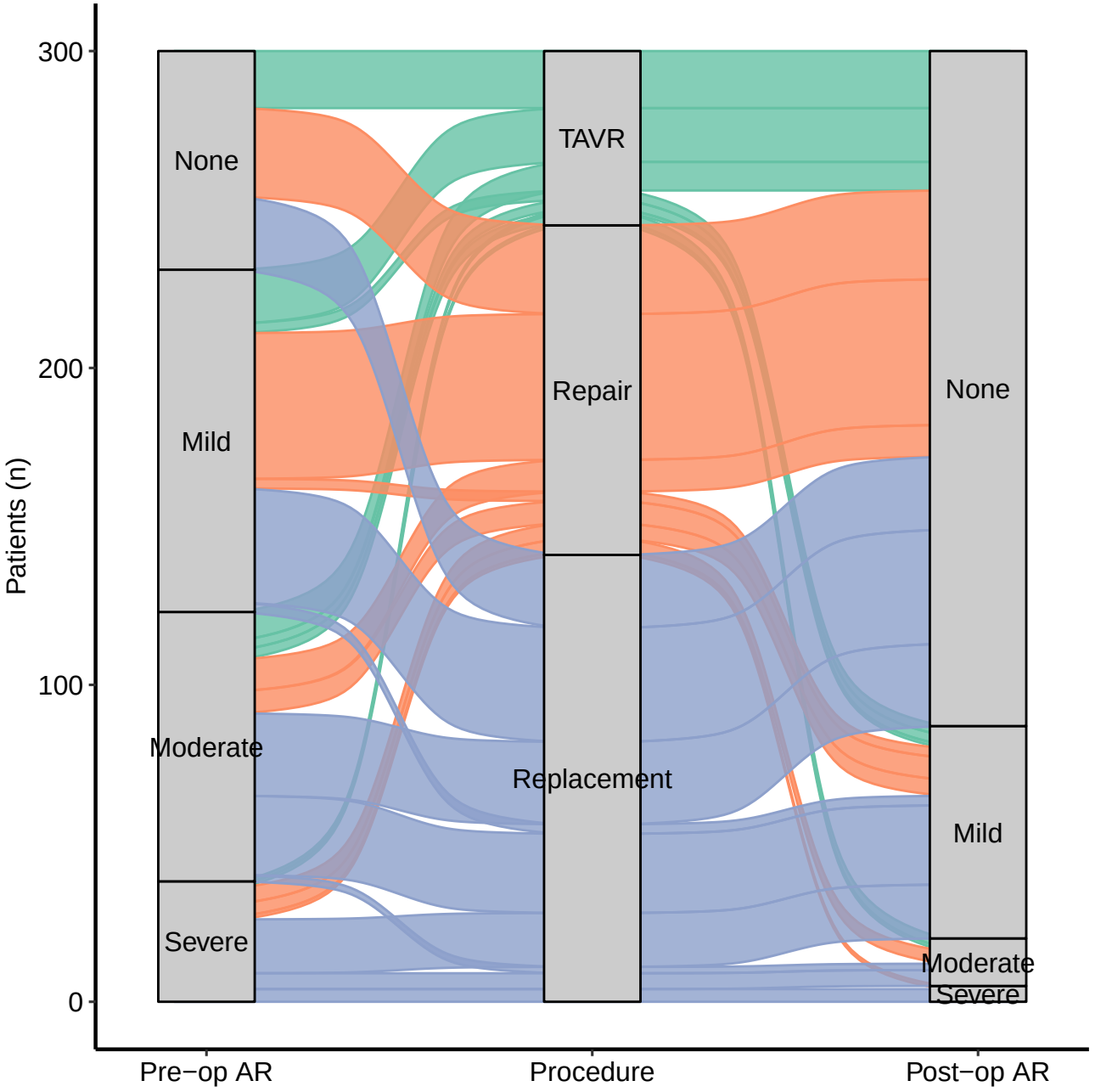

15%
10%
5%
0%
P(AR grade >= 3)
0
2
4
6
8
10
12
Years

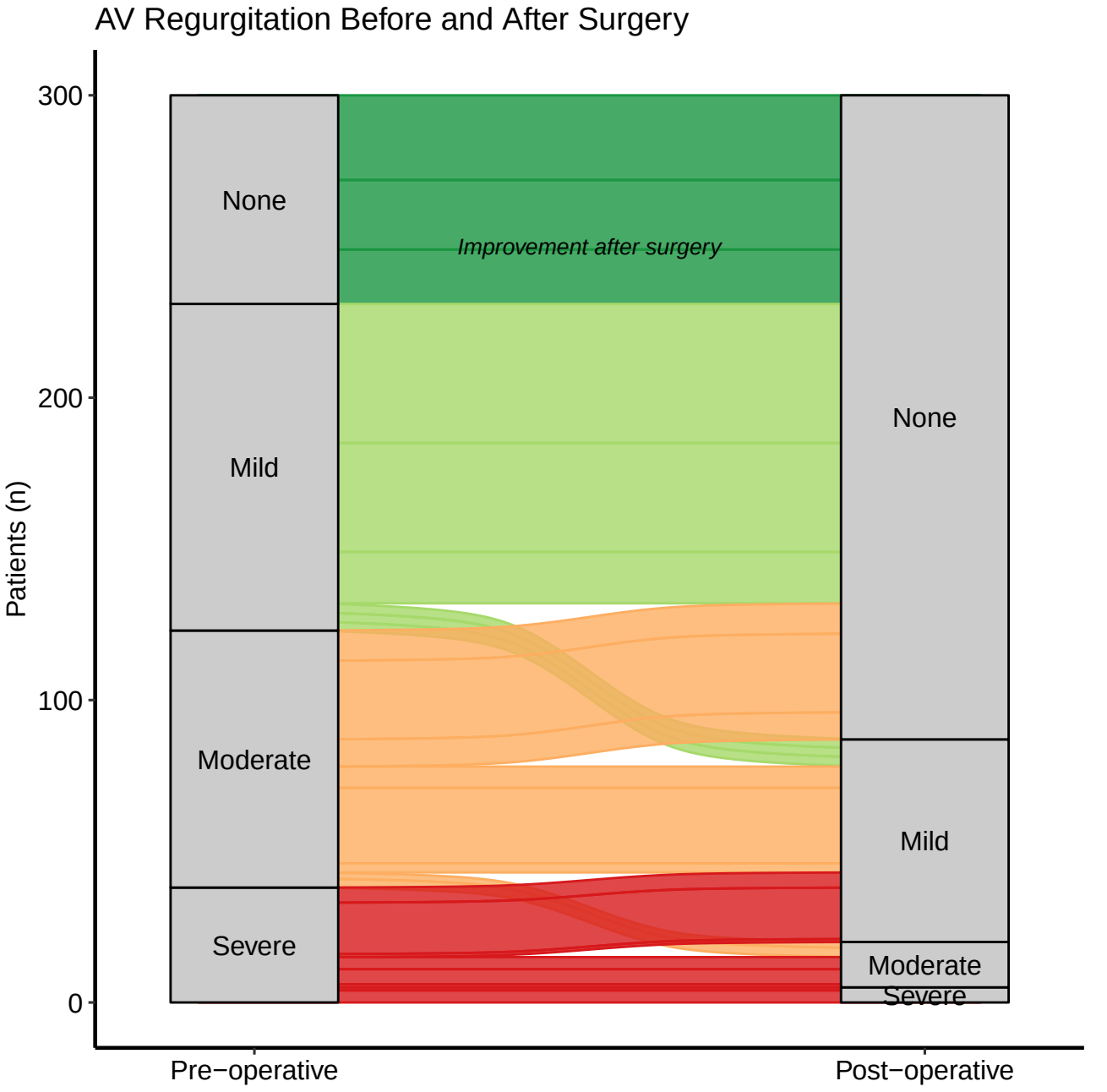

Follow−up (years)
30
20
10
0
1990
2000
2010
2020
Operation Date
Alive
Dead

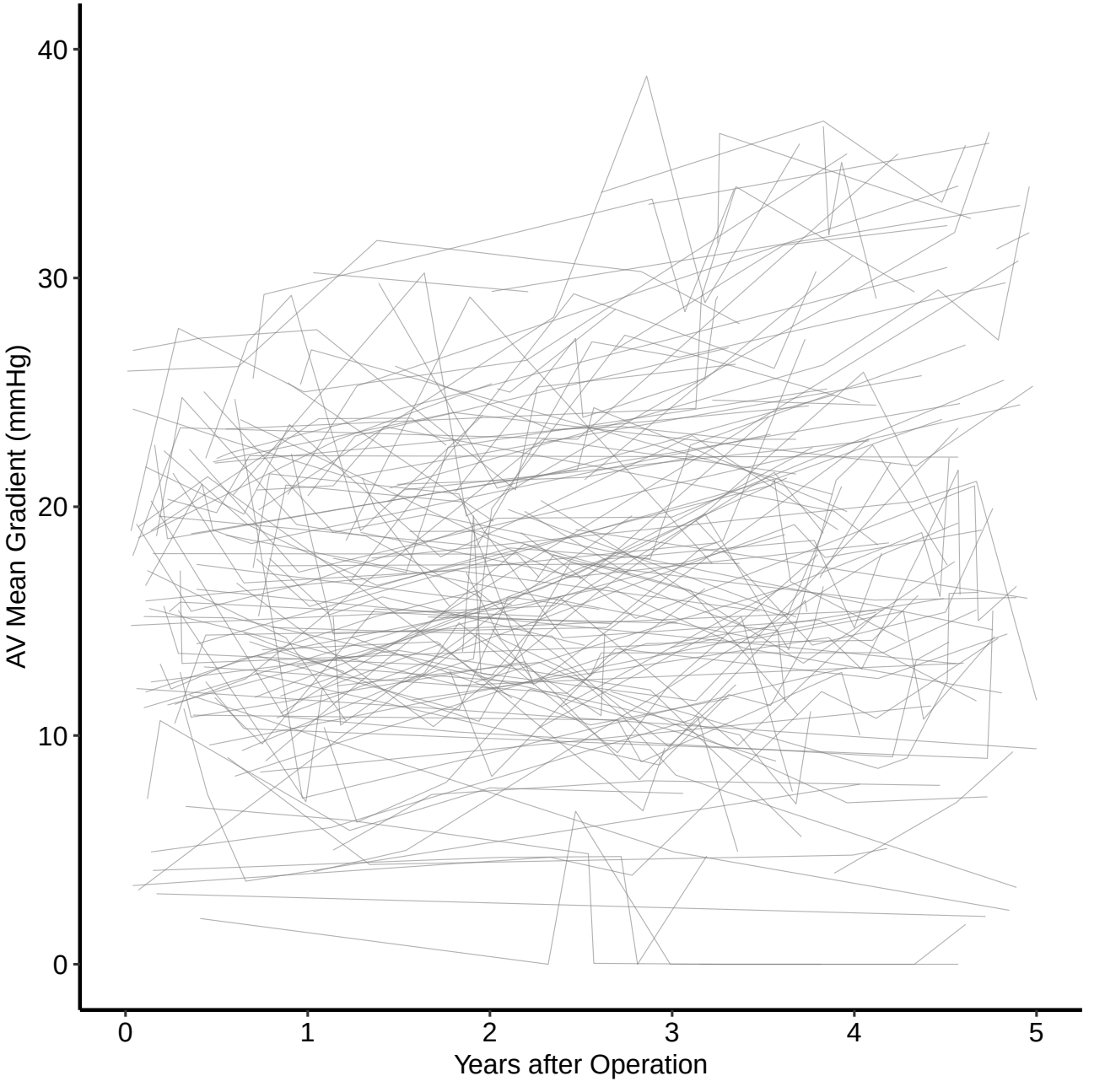

Follow−up (years)
30
20
10
0
1990
2000
2010
2020
Operation Date
No event
Relapse
Death

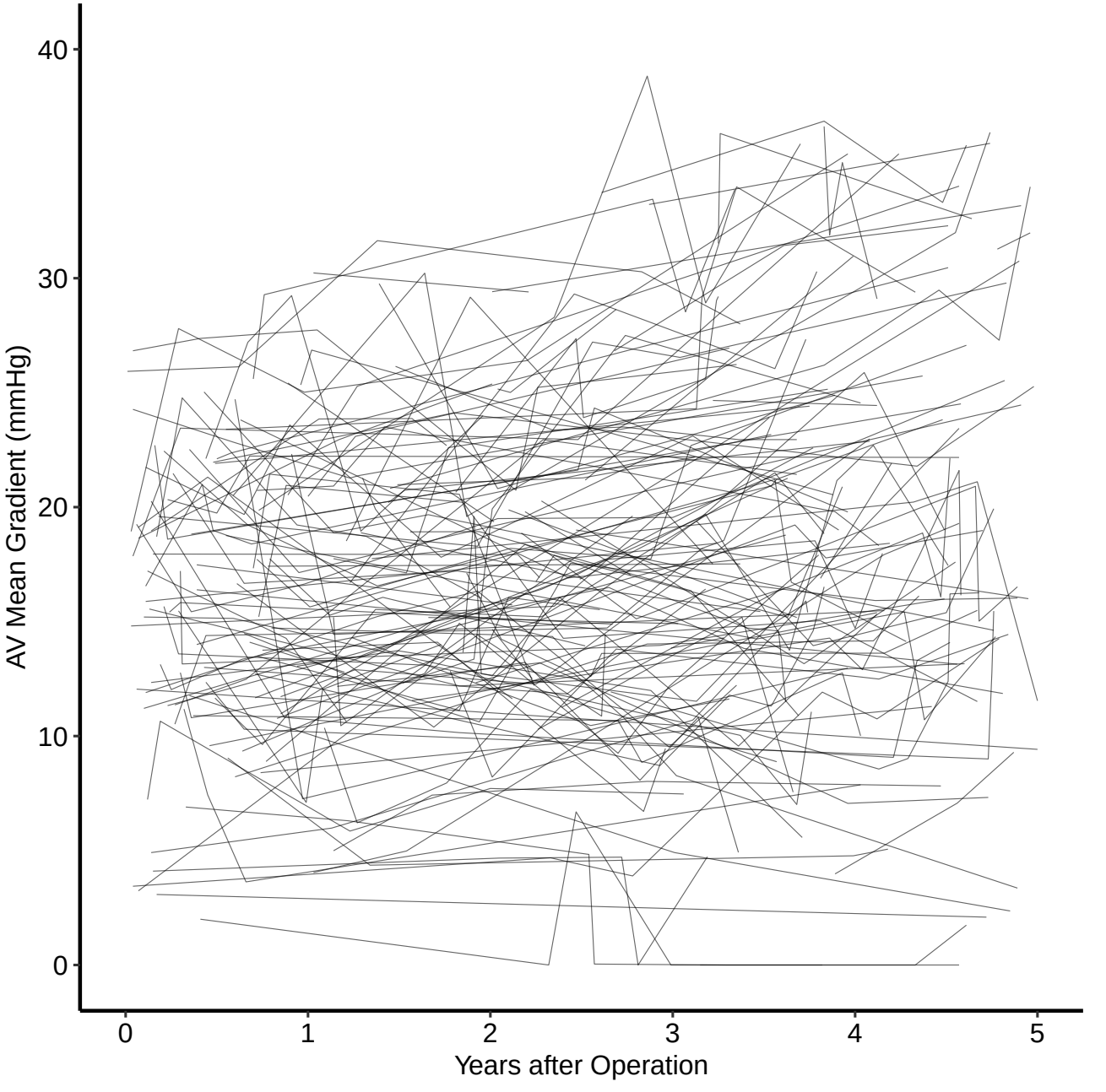

300
200
100
0
freq
None
Mild
Moderate
Severe
TAVR
Repair
Replacement
None
Mild
Moderate
Severe
pre_ar
procedure
post_ar

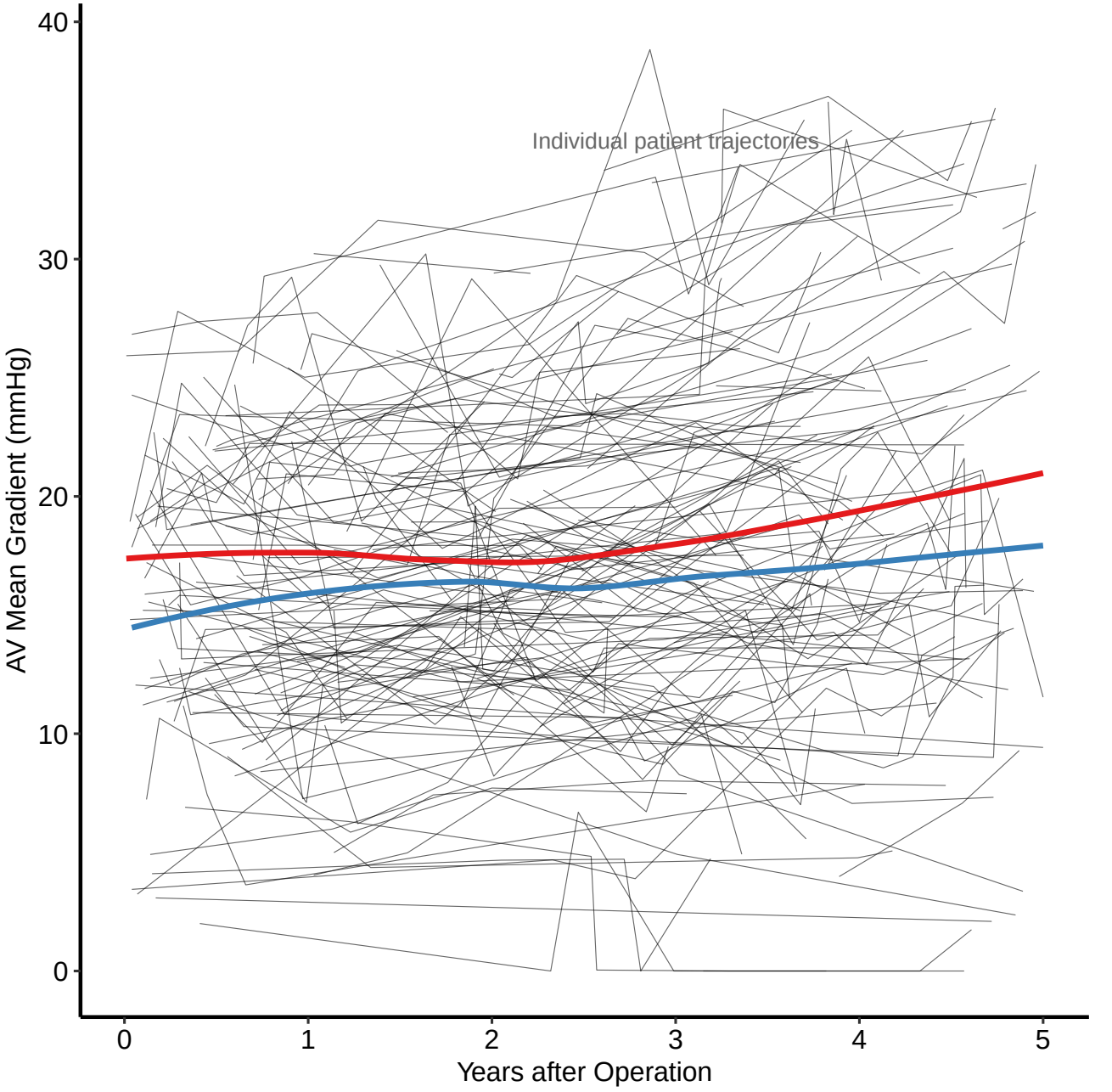

AV Regurgitation: Pre− to Post−operative
Patients (n)
300
200
100
0
None
Mild
Moderate
Severe
TAVR
Repair
Replacement
None
Mild
Moderate
Severe
Pre−op AR
Procedure
Post−op AR

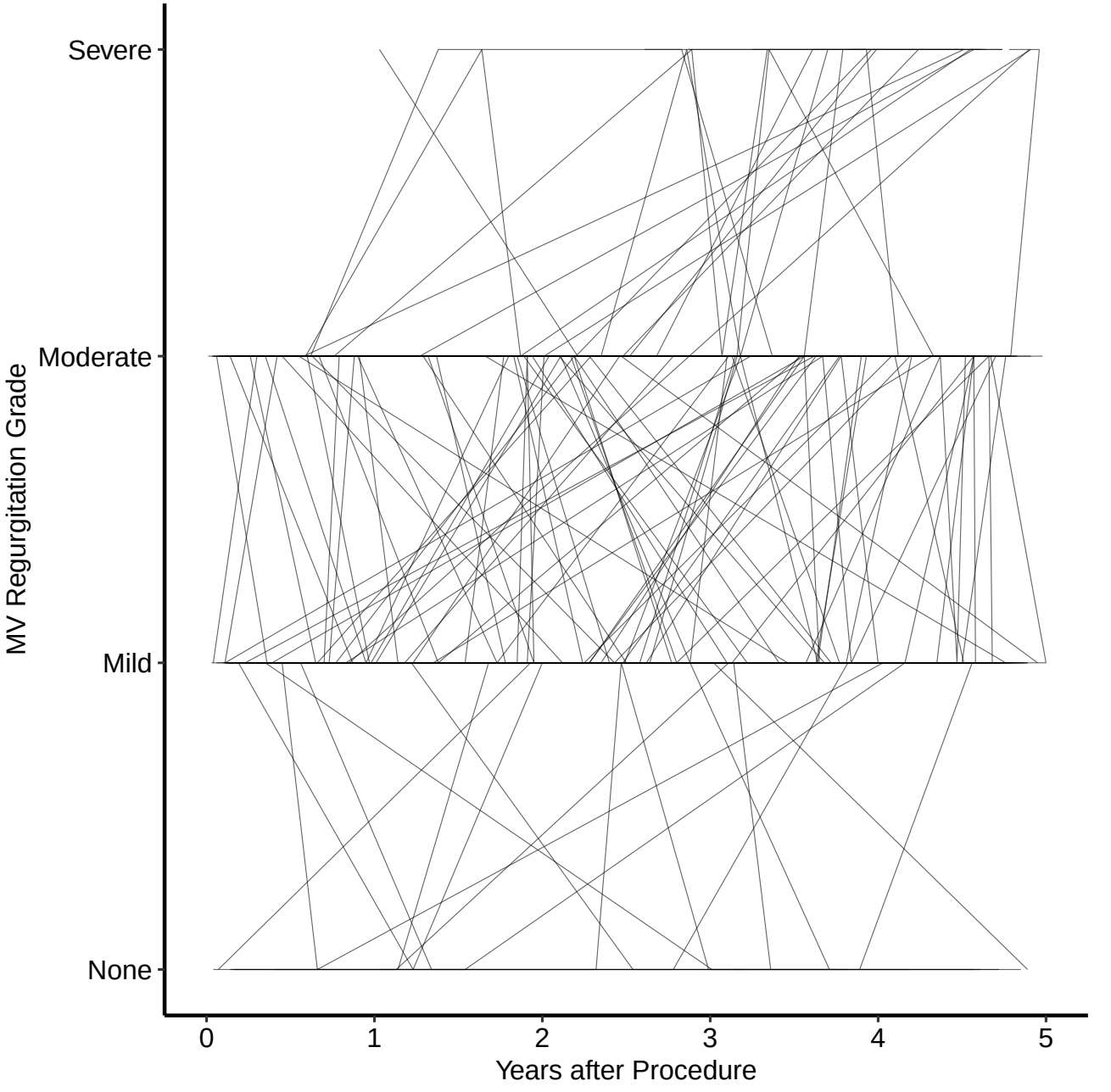

300
200
100
0
Patients (n)
None
Mild
Moderate
Severe
TAVR
Repair
Replacement
None
Mild
Moderate
Severe
Pre−op AR
Procedure
Post−op AR

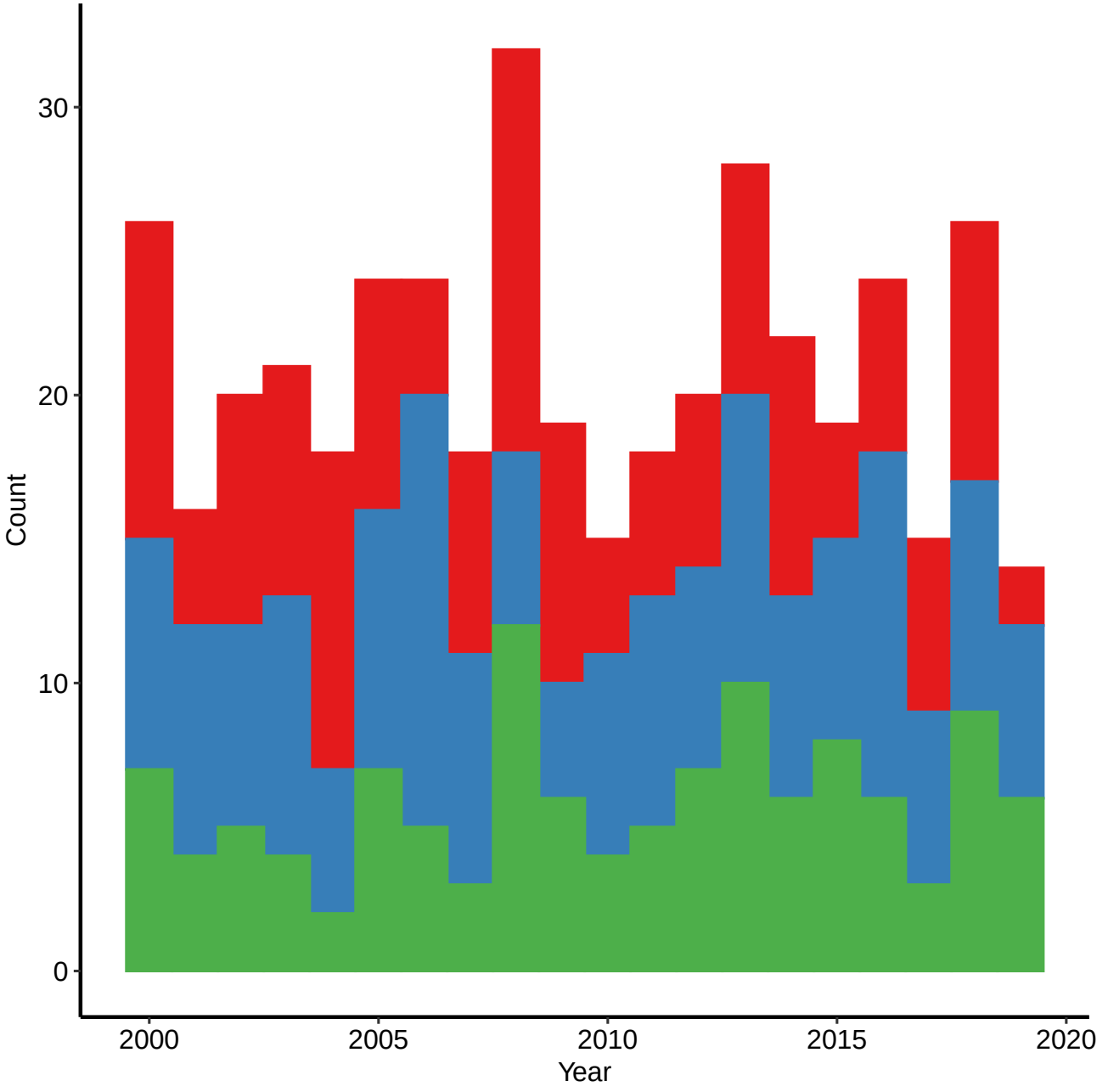

AV Regurgitation Before and After Surgery
Patients (n)
300
200
100
0
None
Mild
Moderate
Severe
Improvement after surgery
None
Mild
Moderate
Severe
Pre−operative
Post−operative

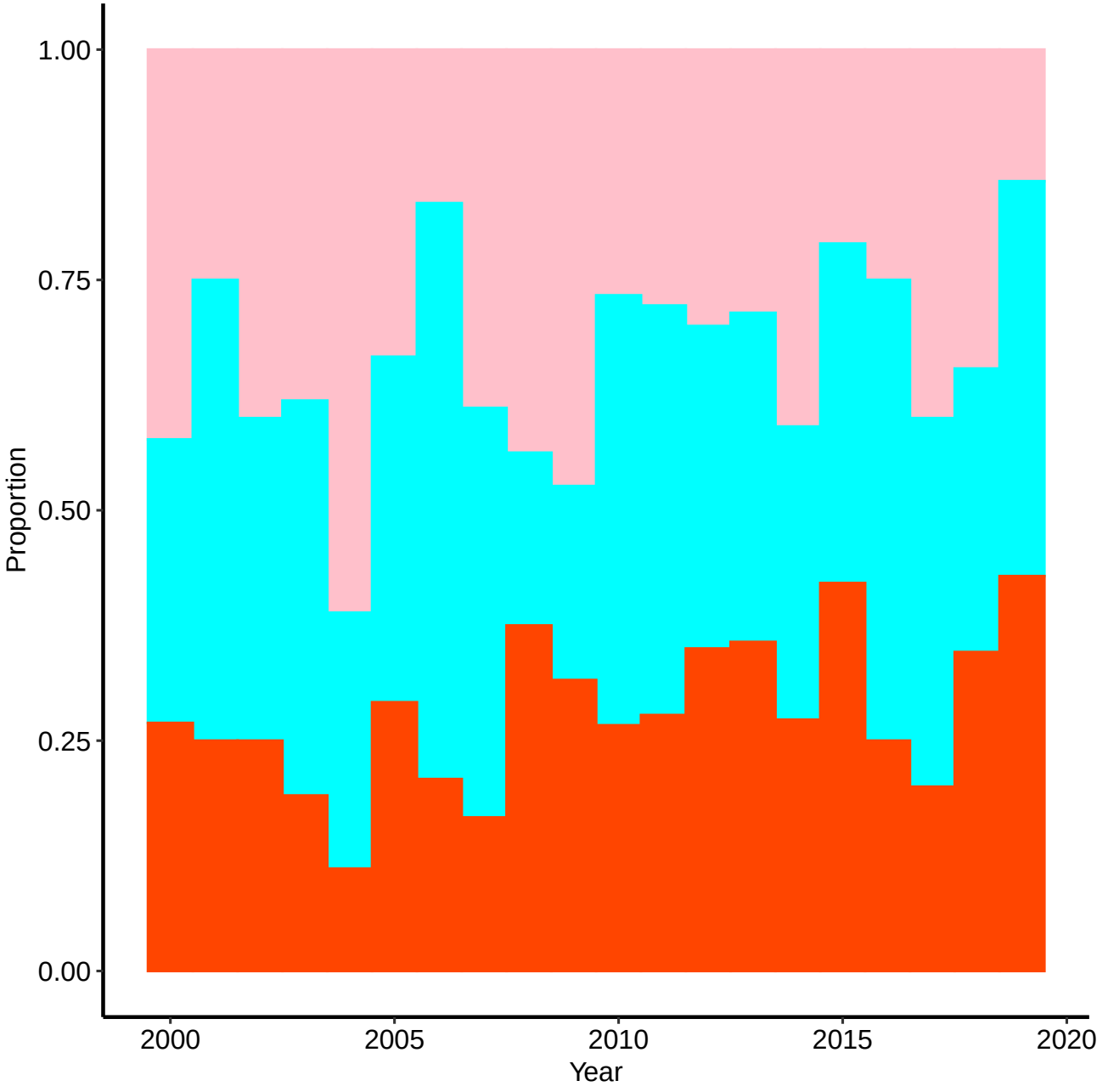

AV Mean Gradient (mmHg)
40
30
20
10
0
0
1
2
3
4
5
Years after Operation

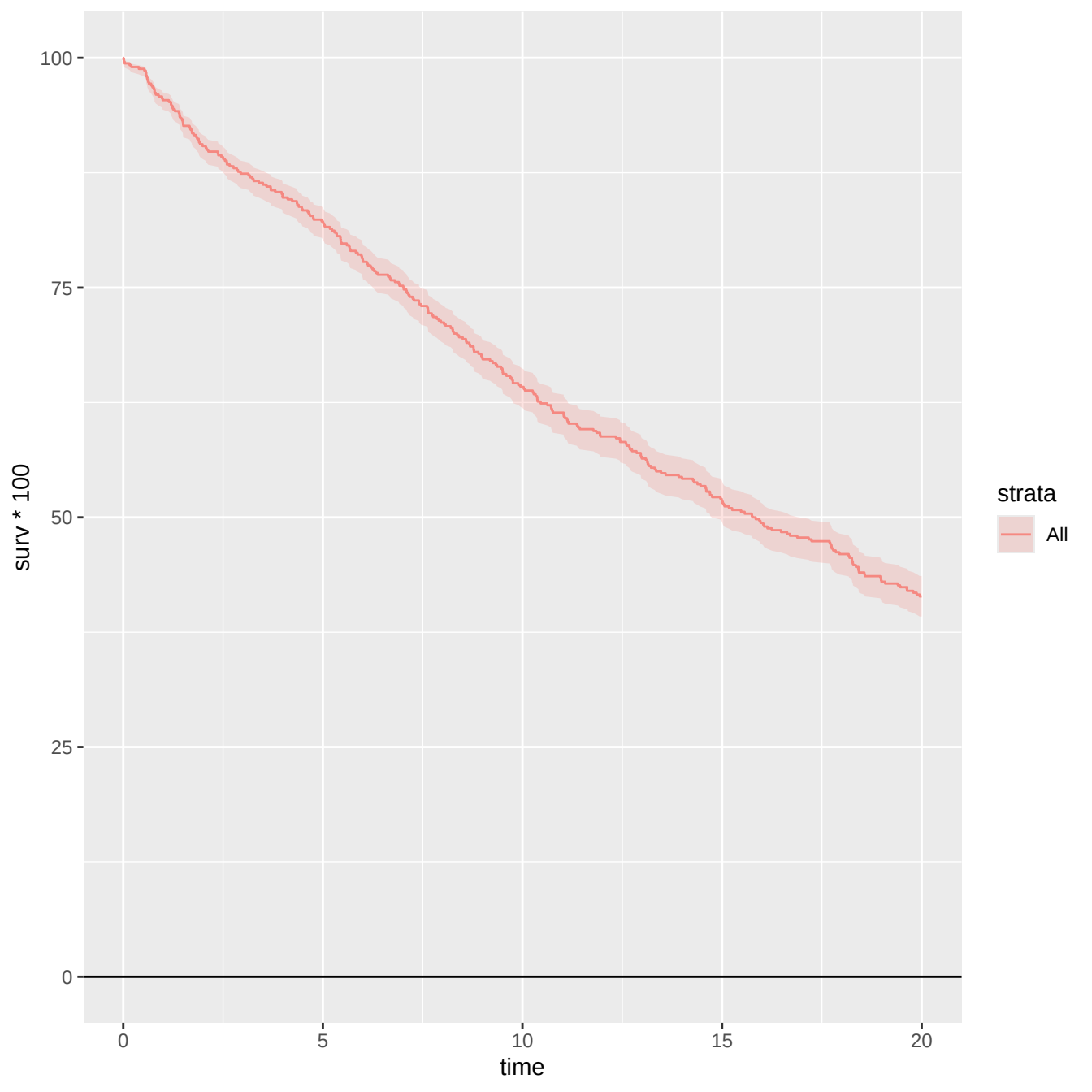

AV Mean Gradient (mmHg)
40
30
20
10
0
0
1
2
3
4
5
Years after Operation

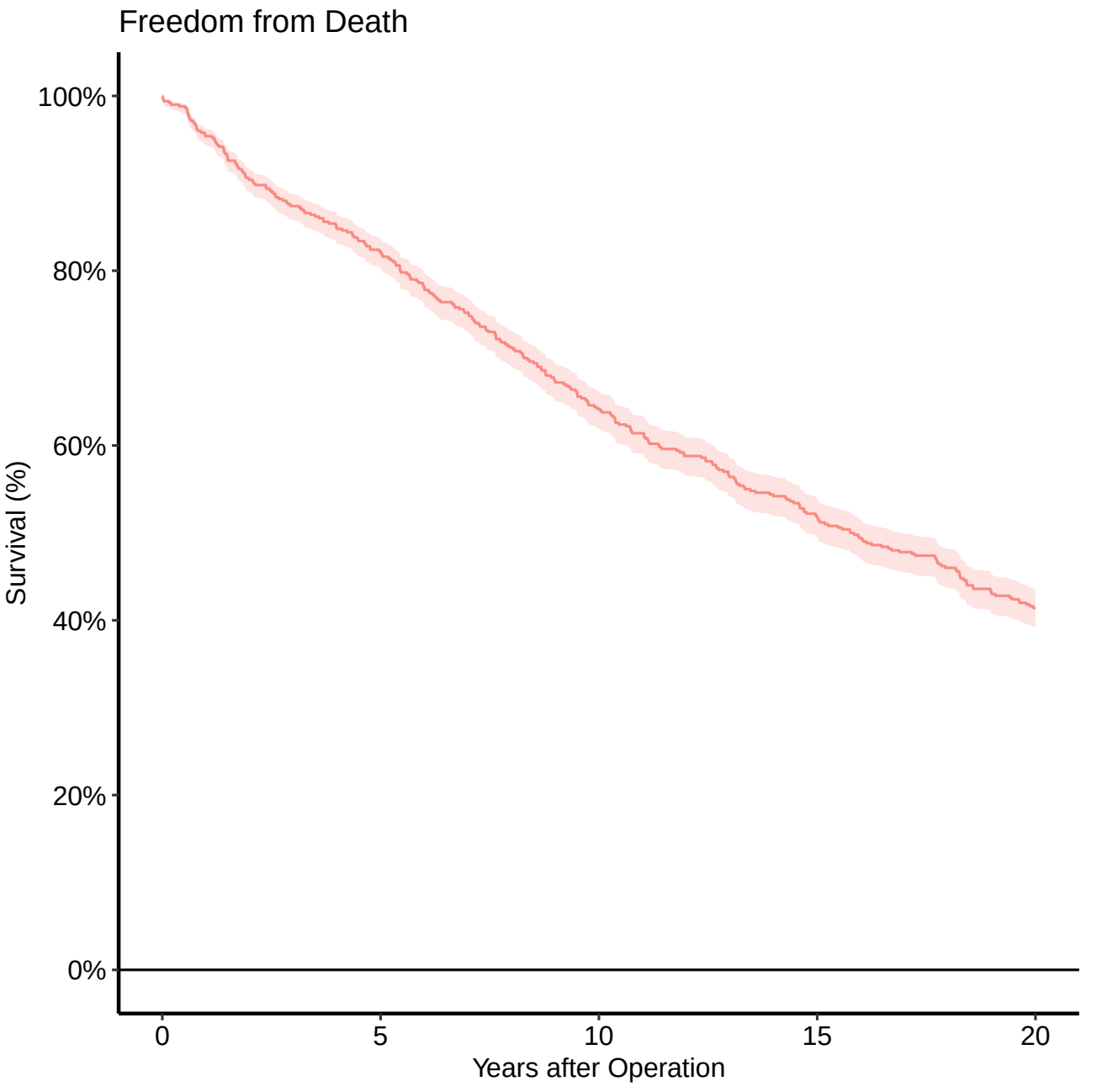

Individual patient trajectories
AV Mean Gradient (mmHg)
40
30
20
10
0
0
1
2
3
4
5
Years after Operation

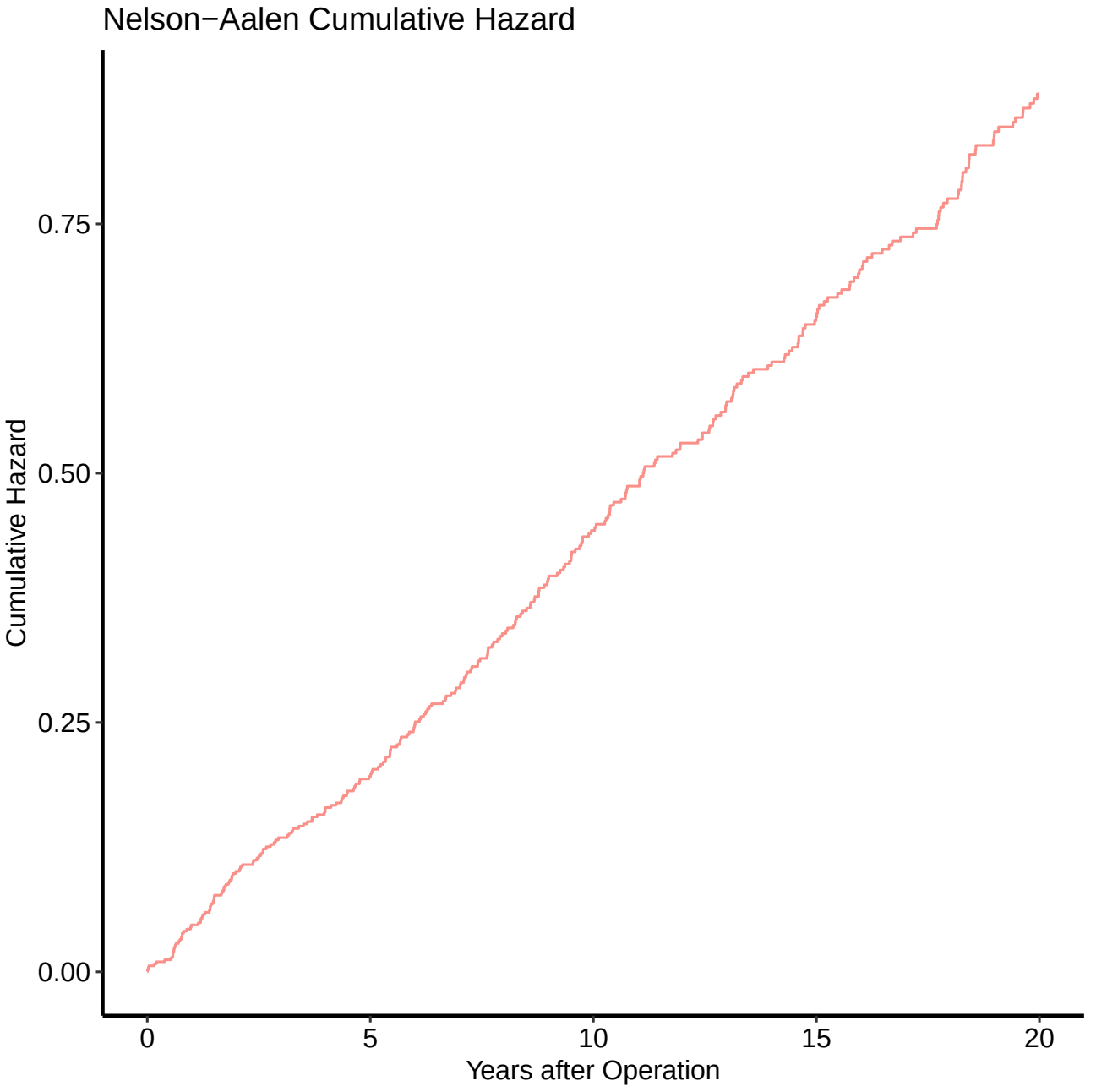

Severe
Moderate
Mild
None
MV Regurgitation Grade
0
1
2
3
4
5
Years after Procedure

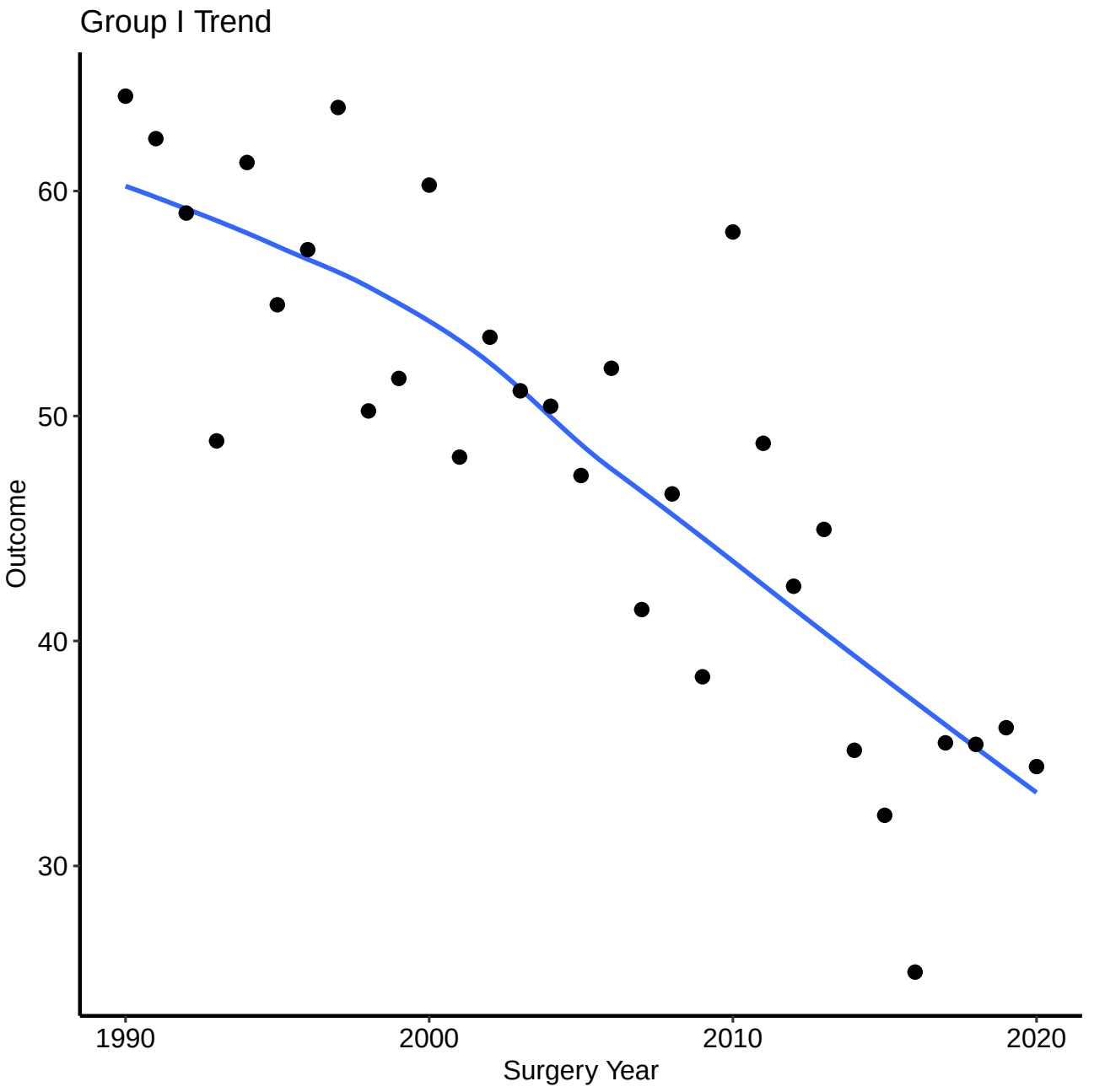

30
20
10
0
Count
2000
2005
2010
2015
2020
Year

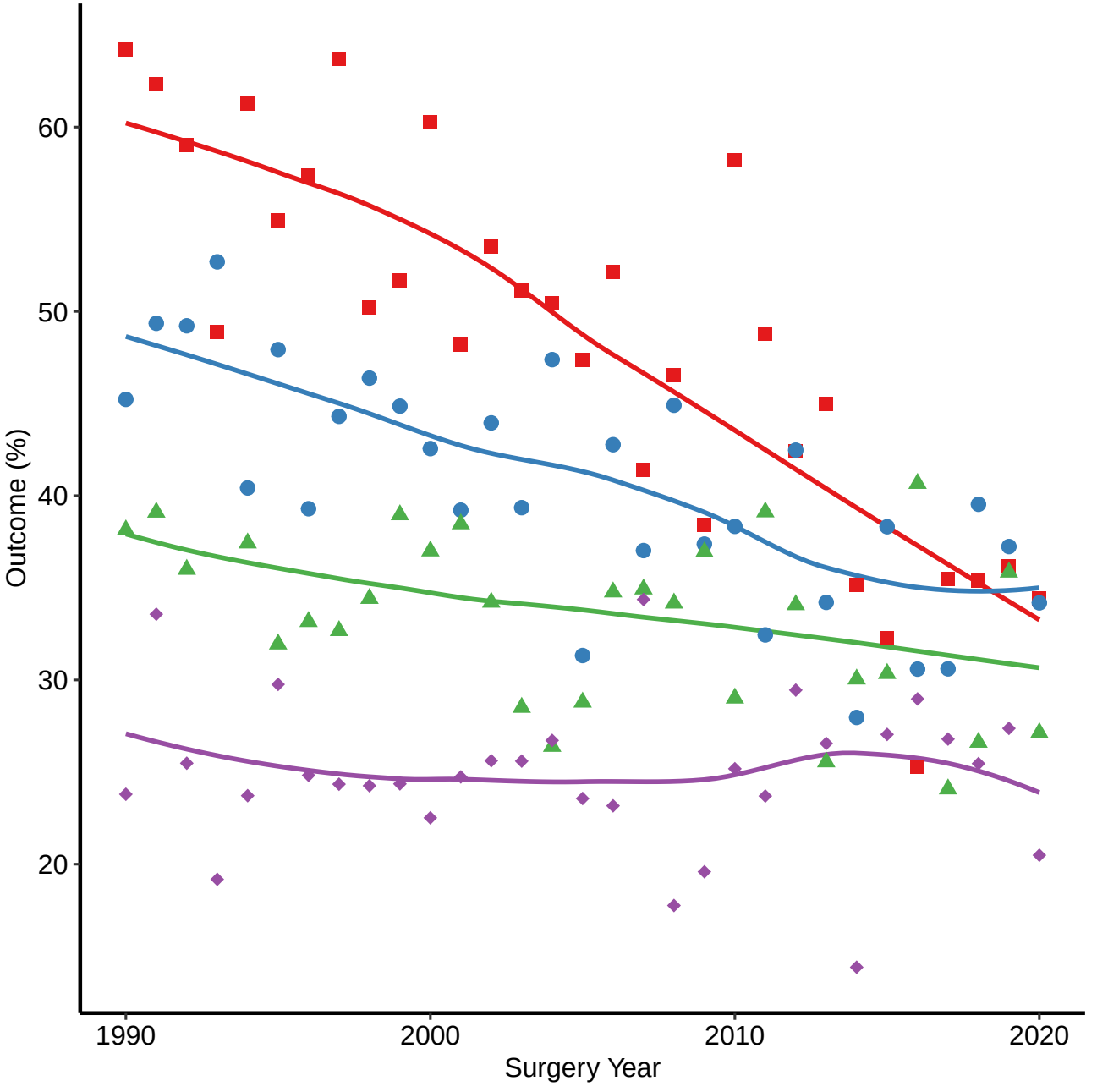

1.00
0.75
0.50
0.25
0.00
Proportion
2000
2005
2010
2015
2020
Year

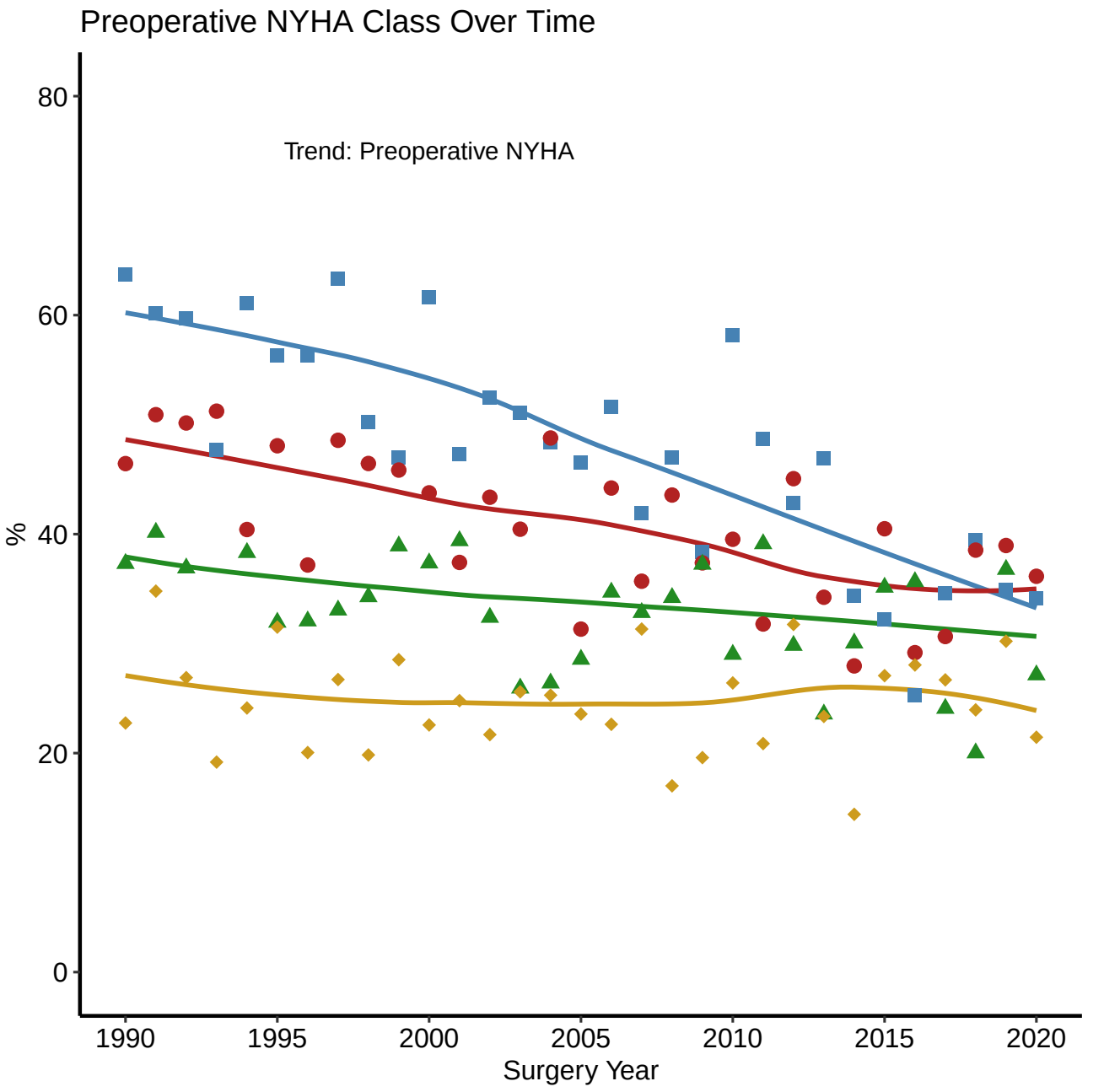

100
75
50
25
0
surv * 100
0
5
10
15
20
time
strata
All

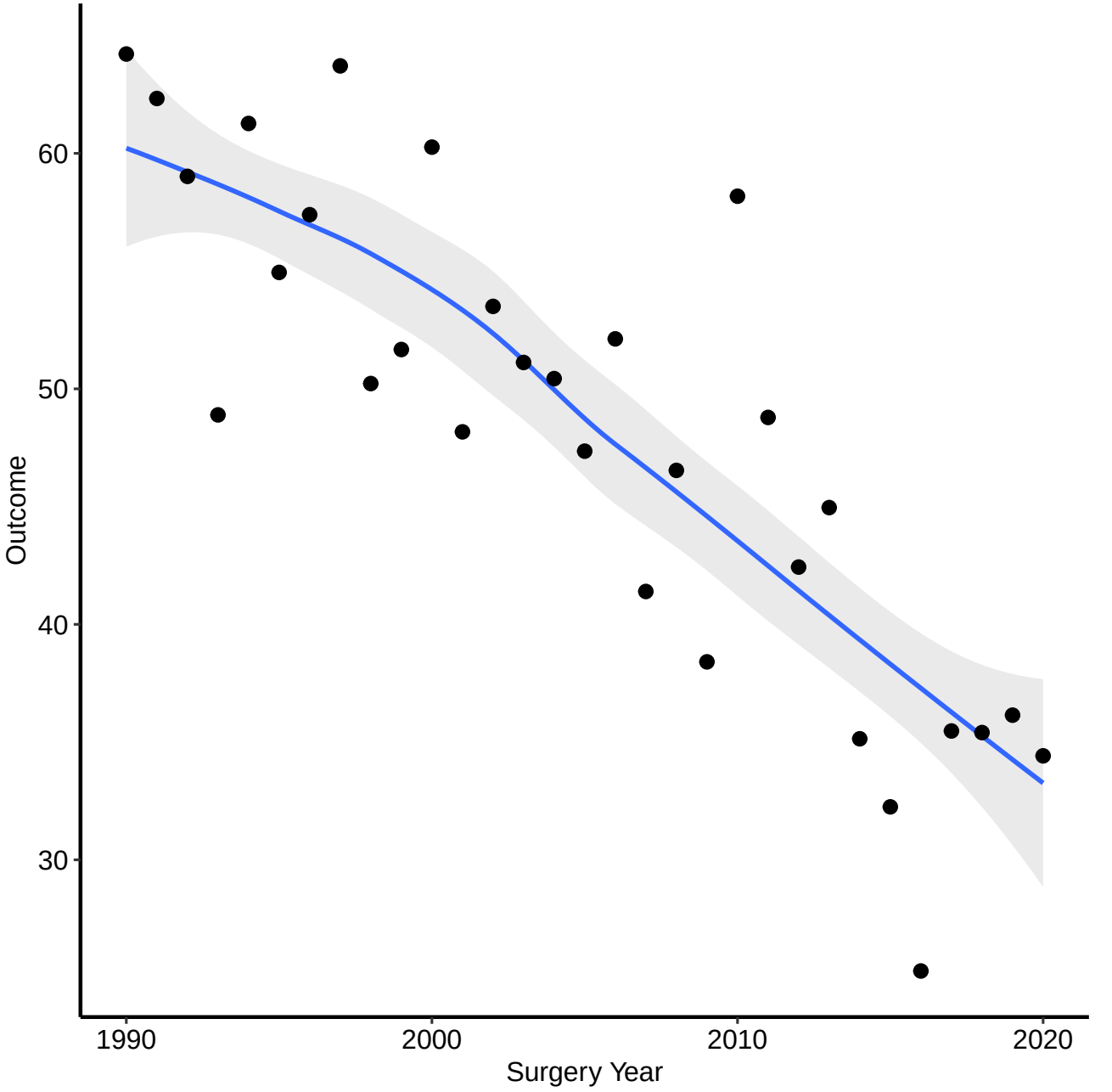

Freedom from Death
100%
80%
60%
40%
20%
0%
Survival (%)
0
5
10
15
20
Years after Operation


Nelson−Aalen Cumulative Hazard
Cumulative Hazard
0.00
0.25
0.50
0.75
0
5
10
15
20
Years after Operation


Group I Trend
Outcome
60
50
40
30
1990
2000
2010
2020
Surgery Year


Outcome (%)
60
50
40
30
20
1990
2000
2010
2020
Surgery Year


Preoperative NYHA Class Over Time
Trend: Preoperative NYHA
%
80
60
40
20
0
1990
1995
2000
2005
2010
2015
2020
Surgery Year


60
50
40
30
Outcome
1990
2000
2010
2020
Surgery Year